American Economic Journal: Economic Policy 2015, 7(2): 88–124
http://dx.doi.org/10.1257/pol.20120137

Can Tax Breaks Beat Geography? Lessons from the French Enterprise Zone Experience†

By Anthony Briant, Miren Lafourcade, and Benoît Schmutz*

This paper shows that urban geography matters to the effectiveness of place-based policies, using the French enterprise zone program as a case study. Whereas this program created more jobs in spatially integrated neighborhoods, its impact on local wages was only visible in the more isolated ones. In addition, a focus on the average impact of the program would lead to the conclusion that it mostly succeeded in displacing preexisting firms, but a lower level of spatial isolation was a clear determinant of the decision to create new firms from scratch. (*JEL* H25, H32, L52, R32, R38)

The evaluation of enterprise zones (hereafter, EZ) has generated a lot of research since the early 1990s, with conflicting conclusions regarding their effectiveness. Arguably, this lack of consensus has paved the way for the current critique vis-à-vis place-based policies (Glaeser and Gottlieb 2008). However, the reasons why some programs work somewhere while other apparently similar programs fail elsewhere has seldom been investigated. We argue that one important component of this heterogeneity is urban geography. The location of targeted neighborhoods within cities, which determines the ease at which people circulate into and out of those neighborhoods, and ultimately, the elasticity of the local labor supply, does matter to reap benefits from EZ programs. If spatial constraints are so high that workers cannot adjust their behavior and commute in isolated neighborhoods, these programs may be efficient at raising earnings among an existing depressed population, but they will not be able to attract firms and create jobs in remote urban enclaves. On the other hand, if the main political issues in these areas are the high level of unemployment and the low firm density, policymakers will only be able to address them in the

* Briant: Paris School of Economics, 48 Bd Jourdan 75014 Paris France (email: anthony.briant@m4x.org); Lafourcade: University Paris-Sud (RITM) and Paris School of Economics, 48 Bd Jourdan 75014 Paris France (email: miren.lafourcade@ens.fr); Schmutz: Howard University, 2400 Sixth Street, Washington, DC 20059 (email: benoit.schmutz@howard.edu). We thank two referees for insightful comments. This research was partly funded by the Direction de l'Animation de la Recherche, des Etudes et des Statistiques (DARES, French Ministry of Labor) and the Secrétariat Général du Comité Interministériel des Villes (SG-CIV, French Ministry of Urban Affairs). We thank discussants from these two institutions, especially Rozenn Desplatz, Dominique Goux, Francoise Maurel, Sébastien Roux and Patrick Sillard. Helpful comments have come from Matthew Freedman, Laurent Gobillon, Xavier d'Haultfœuille, Thomas Piketty, and José de Sousa. This work benefited from the National Agency for Research, through the program "Investissements d'avenir," with the following reference: ANR-10-EQPX-17 (Remote Access to data - CASD). Schmutz thanks the University Paris-Sud and the Paris region for funding, and Georgetown University for its hospitality. Finally, Dorian Jones provided excellent research assistance.

† Go to http://dx.doi.org/10.1257/pol.20120137 to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

accessible neighborhoods but the deadweight loss associated with the policy will also be larger. In this paper, we use the French program "Zones Franches Urbaines" (hereafter, ZFU) as a case study to show that this trade-off actually exists and can be assessed using objective topographic measures of spatial isolation.

Thanks to a series of original indicators of spatial isolation for treated and non-treated neighborhoods, we show that geographical heterogeneity determines the ability of the ZFU program to impact firms' settlements, job creation, and earnings. In particular, whereas a focus on the average impact of the program would lead to the conclusion that it mostly succeeded in displacing preexisting firms, spatial integration was a clear determinant of the decision to create new firms from scratch. Similarly, whereas the program created more jobs in spatially-integrated neighborhoods, its impact on local wages was only visible in the more isolated ones. The demonstration of these heterogeneous effects is our main contribution to the longstanding empirical debate on the effectiveness of EZ programs. From a policy-oriented perspective, we aspire to help establish new criteria for zone qualification. Finally, we also raise the issue of the transferability and scalability of EZ programs, which may turn out to be pretty low if geography, which is not easily modified in the short run, is an important determinant of their overall effectiveness.

The main challenge of our paper is empirical: how can we measure the spatial isolation of a neighborhood in a quantitatively relevant manner? At the intra-urban scale, the location of a neighborhood may not be well proxied by a mere combination of relative distance to business districts or employment centers. It also depends on access to transportation infrastructure as well as on physical elements that might create urban severance—natural obstacles, industrial wastelands or even, paradoxically, large transportation infrastructures such as highways or airports, which are often irrelevant to poor residents. The impact of physical obstacles on the functioning of the city has long been acknowledged by urban planners and geographers. Jacobs (1961) was already pointing out that monofunctional enclaves and large transportation infrastructures were increasingly leading to a new form of city-carving and posed a threat to urban cohesion, in particular if the neighborhoods were small and not diversified enough. Five decades later, however, the quantitative research on this matter is still sparse because the measure of urban severance remains an empirical challenge, compounded by the ambiguous role of transportation infrastructures and the necessity to take the mobility of residents into account.[1] We argue that the level of spatial isolation of an urban neighborhood can only be properly measured as the combination of three essential features: *centrality*, *accessibility,* and *continuity* of the urban landscape. Centrality is a measure of the relative position of the neighborhood with respect to the other locations of the city, and more particularly central business districts. Accessibility depends on the access to the transportation network or nodes connecting the neighborhood to these locations. As for continuity, it can be

[1] As noted by Handy (2003), "whether transportation facilities will serve as borders, barriers, or gathering spots depends in part on how residents perceive and react to these facilities." Similarly, Button (2010) explains that the best indicator of urban severance would include the number of suppressed trips induced by the obstacles, which requires the use of subjective individual surveys and is more in line with a monographic approach of the question.

characterized in reference to the number, the nature and the magnitude of the urban cutoffs which physically isolate the neighborhood from the surrounding locations and generate an urban enclave: we define as *urban severance* this corresponding lack of continuity.

We estimate a series of augmented difference-in-differences (DD, hereafter) models to compare the treated ZFU to a control group formed by similar areas that filled all the eligibility criteria but were not selected into the program. To understand the differentiating role of geography, we interact the treatment indicator with four different indicators of spatial isolation. Our estimates, which can be interpreted as in a triple-difference framework, indicate that spatial isolation does matter to explain spatial differentials in job and establishment creation or transfer rates across ZFU. For instance, during the two years following the implementation of the policy, the establishment settlement growth rate in the areas designated ZFU was, on average, 8.5 percentage points above the same growth rate in the unselected similar areas. However, this result is entirely driven by the less isolated ZFU: splitting the sample of neighborhoods in half according to a global index of spatial isolation, one can see that the less isolated half of the population of ZFU witnessed a 16 percent increase in the growth rate of establishment inflows, while the other half did not see its situation improve at all relative to similar unselected areas. This result is robust to the consideration of several concerns regarding the identification of the model. It can also be refined by investigating how spatial isolation interacts with the program effect for sectors with different capital/labor ratios and different requirements regarding the mobility of these factors.

The rest of the paper is organized as follows: Section I introduces the ZFU program; Section II describes our measures of spatial isolation at the neighborhood level; Section III explains our empirical strategy to evaluate the effectiveness of the ZFU program on different economic variables, and the role played by spatial isolation in this respect; Section IV presents and discusses our results, while Section V concludes.

I. Elements of Context

The ZFU program has become one of the most prominent place-based policies in France. In this section, we briefly document the history of this program, describe its content in terms of tax breaks and discuss the existing evidence regarding its effectiveness and its possible relationships with urban geography.

A. *The ZFU Program*

In the past four decades, spatial disparities have peaked within French cities and their dramatic consequences in terms of segregation, exclusion and eventually juvenile delinquency and violence, have constantly called for innovative political responses. The 1981 program of integrated community development, "Développement Social des Quartiers" (Social Community Development), had been met with little success and in 1996, an even more comprehensive set of measures entitled "Pacte de Relance pour la Ville" (Urban Stimulus Package), was

implemented. Like other concomitant programs around the world, it was designed to generate a big push in favor of the most distressed areas.[2]

The EZ program ZFU was the most ambitious component of this package. It was based on the ultimate level of a three-tier zoning system of deprived neighborhoods: the first-tier level, composed of 751 Urban Sensitive Zones ("Zones Urbaines Sensibles" or ZUS) was initially formed by urban neighborhoods with a derelict housing stock and a low job-to-resident ratio. Among them, 416 Urban Revitalization Zones ("Zones de Redynamisation Urbaines" or ZRU) were more carefully targeted. The selection of these new zones was supposed to stem from their respective ranking according to a synthetic index aggregating the total population of the area, its unemployment rate, its proportion of residents with no diploma, its share of residents under the age of 25, and the tax potential of the hosting municipality.[3] Finally, the 44 ZRU that seemed to be the most underprivileged were declared ZFU (subsequently, these would be known as the first-generation of ZFU or ZFU1G). Firms that entered a ZRU or a ZFU could benefit from various tax breaks and other social exemptions, yet, as detailed below, the generosity of these rebates was much higher in ZFU. Moreover, whereas these rebates were designed to be limited in time, they were postponed in practice, so that all ZFU were still active in 2014, 17 years after the creation of the ZFU1G.

A second wave of 41 ZFU (hereafter, ZFU2G) was created in 2004, out of the stock of ZRU that had not been designated ZFU1G. Whereas this second selection was officially supposed to be based on objective social and demographic criteria, it was mostly driven by the political desire to achieve a more even scattering of the ZFU across France. Givord, Rathelot, and Sillard (2013) show that, depending on the criterion, ZFU2G could be depicted as better-off, worse-off, or very similar to the other ZRU. In the sense that political considerations came into play, the selection of the ZFU2G out of the remaining ZRU was close to random, which makes a DD approach particularly justified. For this reason, we will focus our evaluation of the ZFU program on this second wave: we consider ZFU2G as the treated group, and the set of out ZRU that had not been designated ZFU1G as the control group. This control group stays unchanged over the period preceding the designation of ZFU2G (1997–2003).

Finally, while the program had never really been evaluated, 15 new ZFU were added in 2006 (ZFU3G). By that time, the annual cost of the program reached more than half a billion euros. The cost of ZFU2G alone amounted to 125 million euros in 2005.[4] The final (and current) 100 ZFU (93 in mainland France: 38 ZFU1G, 41 ZFU2G, and 14 ZFU3G) represent 1.5 million inhabitants (against 4.4 in ZUS, and 2.9 in ZRU), and they are mostly located in large cities (See Figure A1 in

[2] The most notable examples include the Social City program in Germany, the Big City program in the Netherlands, the National Strategy for Neighborhood Renewal in Great-Britain, and the HOPE VI program in the United States.

[3] The tax potential is defined as a theoretical product of local taxes in case the average national rate were applied to the municipality for each of the local rates. Many countries have adopted similar multidimensional indices to target their most distressed neighborhoods. For instance, the British government uses an *Index of Multiple Deprivation* based on the aggregation of seven dimensions.

[4] According to Rathelot and Sillard (2007), this represents an average cost of 31,250 euros by job created or, equivalently, 208,000 euros by firm settlement.

Appendix A for details). All ZFU without exception are part of a metropolitan area (hereafter, MA).[5] Overall, 26 ZFU are located within the Parisian MA, which is peculiar, and nine more are located in the next three largest MA (four in Lyon, two in Marseilles, and three in Lille). Until now, and despite little proof of their success, French politicians across the board do not seem to get enough of ZFU.[6] Given this political reality, one can but hope for a better targeting of the ZFU program, which happens to be the main message of this paper.

The ZFU Tax Package.—In ZUS, local authorities are empowered to decide whether firms can benefit from tax exemptions. However, these exemptions are very limited (to real estate contributions, notably) and the fiscal component of the ZUS part of the program is only marginal. By way of contrast, the ZRU and the ZFU programs offer comprehensive deductions of payroll, corporate, business, and property taxes. However, the overall package is far more generous in ZFU, both in scope, length and eligibility criteria. The main conceptual difference is that only new hires can benefit from tax breaks in ZRU, whereas all employees may be impacted in ZFU. Second, exemptions can last up to three times longer in ZFU: for example, firms may fully be exempted of corporate income tax for five years followed by nine years at decreasing rate, against two and three years respectively in ZRU. Finally, many more firms are eligible in ZFU, because the cap on total exemptions is much higher. As a consequence, the overall cost of the ZFU program is larger than for the ZRU program, even though ZRU are far more numerous than ZFU. For instance, in 1998, the 38 ZFU1G cost 237 million euros, compared to the 234 million for 416 ZRU (for a comprehensive description of the respective tax packages, see Appendix B).

Compared to other similar programs, especially in the United States, the ZFU program is very large, both in terms of range (the entire country) and in terms of the magnitude of the tax rebates. The total cost of State Enterprise Zones, Federal Empowerment Zones, and Federal Enterprise Community programs combined amounted to 1.21 billion dollars in 2006 (Ham et al. 2011), whereas for the same year the total cost of the ZFU program alone amounted to 472 million euros or equivalently, 5.6 million euros by ZFU, 1,800 euros by ZFU worker or 360 dollars by ZFU resident. This may be partly explained by the French tradition of place-based programs and of larger public interventions in general.[7] By way of comparison, in the famous Colorado EZ program, the most notable exemption was a 3 percent tax rebate for investment purposes, which cost 1.25 million dollars a year between 1986 and 2000; in Californian EZ, exemptions amount on average to 240 dollars per

[5] Until 2010, French metropolitan areas ("Aires Urbaines") were defined around a city-pole with more than 5,000 jobs and a group of surrounding municipalities polarized around this pole, in which at least 40 percent of the MA workforce was employed.

[6] In 2009, the government called for a systematic evaluation of the program aimed to help decide whether it should be terminated or not. Four research teams were selected to study different aspects of the program, with an expected completion date in January 2012. However, in July 2011, a political report came out, signed by one of the most vocal proponents of ZFU and, in October 2011, the program was officially renewed for five more years. The change of majority in May 2012 did not modify the general political credo and another report, released in May 2013, argued that the impact of the program was underestimated because the existing evaluations did not take all the relevant dimensions into account, and also recommended that it be extended until December 2017.

[7] For a thorough comparison of French and US urban public policy, see Donzelot, Mével, and Wyvekens (2003).

worker in 2005. In the United Kingdom, the Local Enterprise Growth Initiative was equivalent to a 60-pound annual subvention to each working-age resident.

B. *Literature*

Several mechanisms are likely to prevent EZ programs from having net positive effects, in addition to also seriously complicating their evaluation. First and foremost, one has to bear in mind that place-based policies disrupt complex spatial equilibria and may therefore have serious general equilibrium effects such as capitalization into house prices or unexpected demographic dynamics, both leading to the capture of the subventions by untargeted populations. As shown by Kline (2010), if markets are perfect, these programs will mostly benefit the agents in control of the real estate market if workers or firms are really willing to respond to their inception by switching neighborhoods. Paradoxically enough, even though EZ programs are primarily employment programs, they will be less efficient if they impact local unemployment rates, rather than the level of earnings. On the other hand, Kline and Moretti (2013) show in subsequent work that there is an efficiency rationale for local job creation if labor market imperfections, such as hiring costs, are sufficiently high.[8]

Many other negative effects may come into play as pointed out for instance by Charnoz (2014). A *spillover effect*, which is almost inherent to any place-based program, may create distortions in favor of the targeted locations, at the expense of other locations, in particular their surroundings. If these diversions are strong enough, the overall impact of the program may even end up negative. In addition, this might create an upward bias in the estimation of the impact of the program if these surroundings are targeted as a control group. Another important mechanism is the *windfall effect*, because the most distressed neighborhoods are often targeted by many competing social programs. If it is the case, one may no longer be able to isolate the net impact of the EZ, which can be close to zero, regardless of the magnitude of the program itself. The third effect, sometimes called *mailbox effect*, is even more specific to this type of public intervention: since the criterion to benefit from the program is an address, firms may develop strategies whereby they only open a small office in the neighborhood, but do not really create new activity. This phenomenon could in principle be avoided by a very carefully crafted policy, although in practice, this would require too much monitoring from public authorities. The possible consequence of this effect is to observe a divergence between the impact of the program on firm creation, for example, and employment.[9]

Despite the high level of doubt cast by economic theory upon the potential efficiency of the place-based programs, is it still possible to find a positive impact of EZ in the data? Over the past twenty years, a very large body of empirical research has been conducted to evaluate the impact of US EZ programs, which we will not try to

[8] In addition, this capitalization effect is less likely to be meaningful in France, where a very large share of the real estate market in the targeted neighborhoods is made of public housing.

[9] Finally, it is theoretically possible that designation into EZ will have a negative *stigma effect* on the neighborhood's residents. However, we believe this mechanism to be of second order, since most of these neighborhoods are already stigmatized, regardless of their EZ designation.

summarize here.[10] These studies are almost all based on more of less sophisticated versions of DD estimates or propensity score matching estimates. The evidence is mixed. As stated by Lynch and Zax (2011) or Ham et al. (2011), a majority of these studies fail to reject that the program did not have any effect at all. For instance, among many others, Bondonio and Engberg (2000) and Neumark and Kolko (2010) find no sizeable effects of EZ on employment; on the other hand, O'Keefe (2004), or Ham et al. (2011) do find a positive impact on various social and economic outcomes. The long-term impact of these programs is still partially understudied. As shown by Bondonio and Greenbaum (2007), the null average impact of enterprise zone programs may hide complex dynamics related to firms creation and destruction. A recent study by Busso, Gregory, and Kline (2013) summarizes many of the empirical difficulties that are caused by the existence of general equilibrium effects and convincingly sorts them out, to finally conclude that designation into the first round of the Empowerment Zone program had positive consequences on employment and wages for residents, and that these effects were not counterbalanced by population inflows or increases in the local cost of living. By way of contrast, Freedman (2012) finds that the already modest positive impact of the New Markets Tax Credit on low-income neighborhoods is attributable to changes in the composition of residents rather than to improvements in the welfare of existing residents.

Given the important structural differences between the US labor market and the French labor market, the external validity of these (already conflicting) results is low.[11] The first evaluations of the ZFU program demonstrated the existence of a small, short-run effect on firm and job creation rates and unemployment duration (Rathelot and Sillard 2008; Gobillon, Magnac, and Selod 2012). They concluded that such a weak effect should be partly due to the large spatial heterogeneity in the effectiveness of the program. More recent work on ZFU2G by Mayer, Mayneris, and Py (2012) or Givord, Rathelot, and Sillard (2013) confirm these findings and stress upon the negative spillovers of the program, which led many preexisting firms to relocate in the targeted locations. Other recent studies try to tackle the more difficult initial selection problem into the first wave of ZFU in order to assess the impact of the program on a longer time span. Givord, Quantin, and Trevien (2012) show that firms do seem to quickly react to tax cuts, but that this does not translate into better employment conditions in the longer run, which calls into question the capacity of ZFU to improve the local economy. This conclusion is corroborated by Charnoz (2014), who uses the Labor Force Surveys and does not find any impact of ZFU1G on residents' unemployment rate until the program conditioned payroll deductions to a more stringent local hiring clause in 2002: by that time, the program had a positive impact on job creation for residents and on the quality of these jobs (increase of open-ended relatively to fixed-term contracts).[12]

[10] Careful reviews on the significance and magnitude of estimates include Lynch and Zax (2011), Ham et al. (2011), Busso, Gregory, and Kline (2013) or more recently, Neumark and Simpson (2014).

[11] Of course, EZ programs have been evaluated in other countries besides the United States and France. One recent example is the study by Wang (2013), who evaluates the impact of Chinese Special Economic Zones and finds evidence of both agglomeration economies and wage increases that are not fully offset by increases in the local cost of living.

[12] Behaghel, Lorenceau, and Quantin (2013) evaluate the impact of a French tax credit program similar as the ZFU policy, but targeted at rural areas, and find no impact of the program on job and business creations.

Beyond Average Treatment Effects.—As reviewed by Landers (2006), the weak average impact of most EZ programs has often led evaluators to suspect that it was hiding some kind of heterogeneity, pertaining to factors related to the areas in which EZ are located, as well as to how EZ are implemented and administered. In an influential policy report, Bartik (2004) makes the case for investigating in this direction, which he argues is necessary for policymakers to be able to replicate the features of successful programs and conversely, discourage policies that lead to ineffective or even counterproductive EZ. However, given the lack of experimental variation in program designs, going beyond average effects is not an easy task as it requires more statistical power than what is usually available. Bartik (2004) advocates in favor of complementing statistical analysis with outcome impact surveys and focus groups assessing whether the program affected the behavior of businessmen.

This is the path taken by Kolko and Neumark (2010) who seek to study how effects vary within the California EZ program. In addition to showing that the program was more effective for zones with a low share of manufacturing employment, the authors also make use of their own survey; they observe that EZ are more able to boost employment when local EZ administrators spend more time in marketing and outreach activities and less time helping firms get hiring tax credits. The authors rationalize these counterintuitive results by claiming that they may reflect idiosyncrasies of the California EZ program. Unfortunately, although this conclusion is probably the most honest one, it does not really help policymakers make better-informed decisions. We believe that this otherwise quite interesting study reflects the two shortcomings faced by this line of research. First, since surveys are answered by agents who are part of the process, the information they convey is likely to be biased. As already pointed out by Bartik (2004), firms will always have incentives to claim that assistance, when it takes the form of cash transfers, had an impact on their behavior. Similar caveats are very likely to happen with local politicians or administrators. Second, there must be some kind of theoretical background explaining the possibility of heterogeneous effects. Otherwise, looking for multiple heterogeneous impacts, over a wide range of location characteristics, can quickly turn into data mining and will expose to Deaton's (2010) critique regarding the economic interest of results that are sometimes purely statistical.

In this paper, we look at a fundamentally urban dimension of heterogeneity: the level of geographical isolation of the targeted neighborhoods, which determines the ability of people to circulate between them and the rest of the economy. There are two reasons for doing so. The first is political: spatial isolation of deprived urban neighborhoods, and its corollary regarding the low mobility of their residents, are major obstacles to the successful integration of these populations (see, among dozens of others, Coulson, Laing, and Wang 2001; Zenou 2002; or Patacchini and Zenou 2007). However, even though EZ may be most needed in these neighborhoods, the second reason to look for the impact of geography is that many arguments lead to believe that place-based policies like EZ will be less effective there. First, on firm creation, one should bear in mind that the neighborhoods designated EZ are often too small and poor for firms to sustain if they cannot interact with larger urban markets. Spatial isolation will hinder these necessary interactions, for instance by discouraging potential customers, up to the point where tax breaks cannot get high

enough for the firms to take the bait and locate in the EZ. Predictions regarding the impact of geography on employment are a little less clear: given that residents of isolated neighborhoods may not have other alternatives, firms created in the EZ could be more attractive to them than they usually are, which could impact local unemployment. However, it is difficult to imagine that, except for a few exceptions, the firms would be able to operate on the local workforce only.[13] If they need to attract people from other parts of the city, they will have all the more trouble to do so as the neighborhood is not easily accessible, which may jeopardize their global ability to function, and this latter effect is likely to dominate the impact on residents' employment. Finally, the predictions regarding the impact of isolation on earnings are opposite: if, as we suspect, isolated neighborhoods are not able to benefit from the EZ policy to create many jobs (because, for instance, outsiders are not willing or even able to commute in), the policy will be more likely to increase earnings there.

II. Geography of the ZFU Program

In this section, we explain the construction of our measures of spatial isolation, which combine elements of accessibility of the neighborhood to urban transportation, centrality of the neighborhood within the metropolitan area and continuity of the urban landscape between the neighborhood and the other parts of the metropolitan area. To capture such elements, we make use of the 2006 version of a topographical database called BD-TOPO, developed by the French National Geographical Institute. BD-TOPO summarizes all the landscape elements of the French territory, at a metric accuracy, in particular public infrastructures and their building footprint (such as universities, hospitals or city halls), relief, hydrography, and vegetation. We focus, although not exclusively, on the transportation network, which is described very precisely,[14] and where infrastructures are ranked according to different characteristics. For example, there are six different levels of roads, depending on the intensity of traffic, with a variable indicating whether each of these roads has two separate lanes, in which case it is considered impassable. The geographical referential upon which these series of maps are drawn (Lambert-93) is the same as the one used by policymakers who design ZFU and ZRU. For this reason, the borders of the neighborhoods are perfectly identified with respect to all the information of the BD-TOPO.

Urban Severance.— To measure how separated the ZFU are from the other parts of the MA, we build two types of indicators related to urban severance. The first type is the number of cutoffs that separate the ZFU from the CBD of the MA.[15] We isolate rivers, railroads, or roads and for the latter, we alternatively consider all impassable

[13] In France, surveys indicate that finding local residents who are qualified enough is the most challenging task for many firms located in ZFU. According to Quantin (2012), only 27 percent of the new hires in ZFU2G between 2007 and 2010 were ZFU2G residents.

[14] We can locate harbors, airports, bridges, cable cars and, most importantly for our purpose, roads, highway junctions, railroads, train and metro stations. Unfortunately, the BD-TOPO does not provide information on bus lines.

[15] We use the term "Central Business District" or CBD to refer to a municipality hosting more than 50 percent of either the MA population or the population of the largest inhabited municipality in the MA. Small MA generally

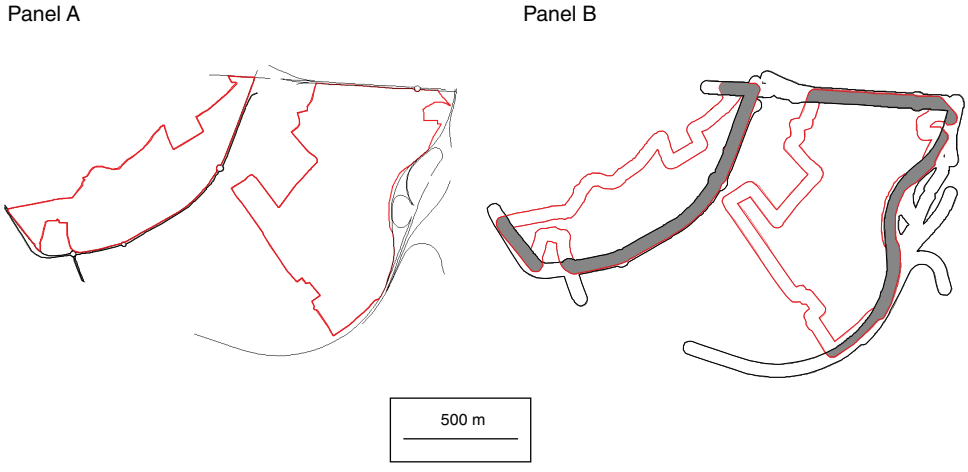


FIGURE 1. SEVERANCE AT THE BORDER OF THE ZFU CAUSED BY VERY HIGH-TRAFFIC ROADS

Notes: (i) ZFU #2315NZF: "La Madeleine" in Évreux; (ii) Panel A: the thick line is the border of the ZFU; thin lines are expressways; (iii) Panel B: the lines are either the border of a 100 meter-wide buffer-zone centered on the ZFU border or the border of a 100 meter-wide buffer-zone centered on the whole set of expressways; the shaded area is the intersection of the ZFU buffer with the road buffer.

Sources: GIS SG-CIV and BD-TOPO

roads (i.e., roads with two separate lanes), or only roads with the highest traffic (i.e., expressways).[16] For MA with several CBD, we compute the average number of obstacles between the ZFU and each CBD, weighted by the share of the CBD in the total population of CBD. The second type of indicator aims at taking into account the fact that some ZFU borders literally follow main traffic arteries, which isolate them from the rest of the agglomeration because they cannot be crossed easily by pedestrians. To account for these "traffic barriers," we draw 100 meter-wide buffers around the railroads and roads located in the vicinity of a ZFU, and we measure the share of a 100 meter-wide buffer around the ZFU border, which intersects with these road buffers. These indicators are built using more exclusive or inclusive definitions of roads. We consider five types of roads: highways (which correspond to the sixth class in BD-TOPO), expressways (fifth and sixth classes), big roads (fourth to sixth class), busy roads (third to sixth class), and impassable roads (any class, but with two separate lanes). The two maps in Figure 1 describe how it is done for the ZFU of Évreux (a city located West of Paris): expressways form 46 percent of the ZFU border, according to our analysis.

Accessibility.—We first compute different indicators of transportation accessibility. The most straightforward indicators are given by the smallest distance between the ZFU border and various transportation nodes, such as highway junctions or train and metro stations. However, we also complement these measures in two ways.

have one CBD only, whereas the largest MA may have several CBD. For example, as depicted in Figure A2 in Appendix A, the Paris MA has 20 CBD, which correspond to the 20 parisian well-known districts.

[16] Expressways correspond to the fifth and sixth classes of roads in BD-TOPO.

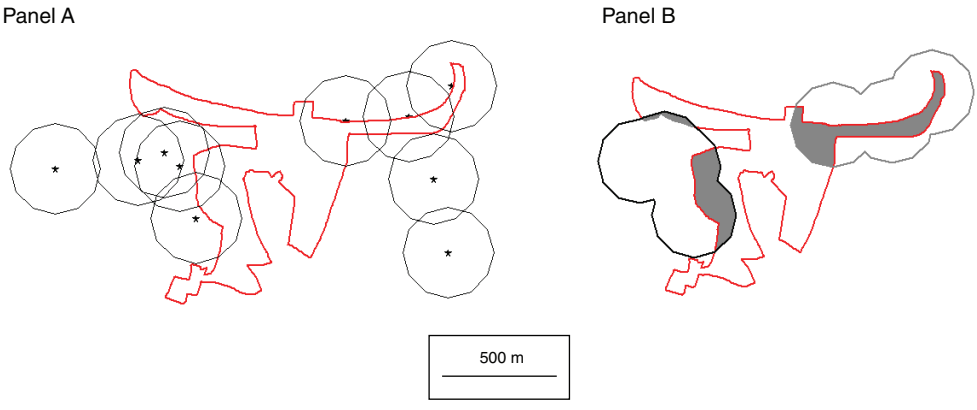


FIGURE 2. THE ACCESSIBILITY OF A ZFU TO TRAIN OR METRO STATIONS

Notes: (i) ZFU #31041ZF: "Faubourg de Béthune-Moulin-Lille Sud-L'Epi de Soil" in Lille; (ii) the thick line is the border of the ZFU; stars are the train or metro stations and circles form a 500 m perimeter around each station; (iii) the shaded area is the part of the ZFU, which is less than 500 m away from a train or a metro station.

Sources: GIS SG-CIV and BD-TOPO

The first is to count the number of transportation nodes in the vicinity of a ZFU and the second, to measure the share of the ZFU which is in the vicinity of such nodes. Figure 2 provides an example extracted from our GIS analysis of a ZFU in Lille, in the north of France. In the left-hand-side map, each of the seven circles intersecting with the thick line indicates one train or metro station which is less than 500 meters away from the ZFU. In the right-hand-side map, the shaded area indicates the part of the ZFU which is less than 500 meters away from a station, which corresponds in our example to 23 percent of the ZFU.

Centrality.—Since these accessibility indicators describe very local situations, one may also want to take into account the broader location of the ZFU within the metropolitan area. ZFU, which can be derelict portions of city centers as well as remote suburbs, vary tremendously in terms of this centrality dimension (see the example provided in Figure A2 in Appendix A for the Parisian metropolitan area). To that end, we build a series of indicators measuring a "Market Potential" of the ZFU à la Harris (1954), based on its distance to all other municipalities within the MA. For each ZFU located in municipality k in urban area "MA," the formula is given by:

$$MP_{ZFU}(x) = \frac{\sum_{k' \in MA, k' \neq k} \frac{x_{k'}}{dist_{(ZFU;k')}}}{\sum_{k' \in MA, k' \neq k} x_{k'}}, \quad (1)$$

where $(k')_{k'=1,\ldots,K}$ are the other municipalities in the MA, $dist_{(ZFU;k')}$ is the distance between the centroid of the ZFU and the centroid of municipality k' and x a measure of the "potential" of this municipality. Variable x can be alternatively the

Table 1—The Geographical Features of ZFU

	Min	Max	Average	SD
Panel A: Urban cutoffs between the neighborhood and the CBD				
Average river cutoffs to all CBD in the MA	0	481.25	34.534	81.675
Average railroad cutoffs to all CBD in the MA	0	38.15	8.579	10.232
Average expressway cutoffs to all CBD in the MA	0	40.85	7.464	8.693
Average impassable road cutoffs to all CBD in the MA	0	41.70	8.459	9.140
Panel B: Severance at the border				
Road severance at the border (busy roads)	0.197	0.913	0.500	0.155
Road severance at the border (big roads)	0	0.749	0.268	0.149
Road severance at the border (expressways)	0	0.498	0.137	0.122
Road severance at the border (highways)	0	0.213	0.025	0.051
Road severance at the border (impassable roads)	0	0.504	0.173	0.114
Road severance at the border (railroads)	0	0.325	0.092	0.093
Panel C: Distance to transportation				
Distance to closest train or metro station	0	2.816	0.739	0.733
Distance to closest highway junction	0	71.712	11.106	14.852
Distance to closest airport	0.227	92.675	21.146	22.227
Panel D: Catchment area				
Share of ZFU less than 500 m away from station	0	0.874	0.123	0.216
Number of stations less than 500 m away from ZFU	0	10	1.355	2.353
Panel E: Market potentials				
Population access in the MA	0.022	0.242	0.091	0.046
Road access in the MA	0.024	0.215	0.082	0.038
Road access (weighted by traffic) in the MA	0.024	0.205	0.084	0.038
Train station access in the MA	0	0.284	0.083	0.051

Notes: (i) The observations are the 93 ZFU in mainland France; (ii) The cutoff variables give the number of cutoffs between the ZFU and each CBD of the MA weighted by the share of the corresponding CBD in the total population of CBD in the MA, divided by the number of CBD; (iii) Road severance is the proportion of the border of the ZFU that is within 100 m of a transportation artery; (iv) Population, road, and train station accesses are market potentials where the variable x in equation (1) is respectively population, length of roads, and number of train or metro stations; (v) Distances are in km; (vi) Panels A and B describe severance indicators, panels C and D describe accessibility indicators, and panel E describes centrality indicators.

Sources: GIS SG-CIV and BD-TOPO

population of the municipality, its number of train/metro stations, or the length of its road network that may or may not be weighted according to the magnitude of traffic documented in BD-TOPO.[17] The centrality indicator is divided by the total sum of x in the MA to mitigate the impact of the largest areas, in particular the Paris region. Such weighted averages of the distance to economic opportunities available in a city are similar to what has been used for many years in the spatial mismatch literature (see for instance Rogers 1997).

Summary Statistics.—Even though they share common characteristics, both in terms of large-scale location and landscape, the neighborhoods that became the 93 ZFU of mainland France are quite diverse with respect to their relationship to their immediate surroundings. For the whole set of indicators, Table 1 describes this diversity.

[17] In which case, the weights go from six for highways and simili-highways to one for small roads and paths.

The heterogeneity is very salient for severance indicators. For instance, the number of river cutoffs separating a ZFU from the CBD of its metropolitan area ranges from 0 to 481. This very large number corresponds to the ZFU of "Val Fourré" in the municipality of "Mantes-La-Jolie" (30 miles northwest of Paris), located along the Seine River that crosses there many of its tributaries. The dispersion is also salient for accessibility indicators. Whereas many ZFU do not count any passenger station in their surroundings, the number of stations within 500 meters of the neighborhood rises up to 10 in the ZFU of "Anzin," located in the city of Valenciennes, where tramway lines form a loop serving the northern part of the city-center. Finally, ZFU located in small metropolitan areas are, in general, much more central than the ZFU located in the suburbs built in the remote periphery of the largest cities to answer the acute housing shortage of the after-war years. An example of the diversity in this third dimension is developed in an online Appendix, which presents two polar examples of ZFU: "les 4,000," located in a notorious northern suburb of Paris called "La Courneuve," and "Chantereigne-Montvilliers" located in the Northern part of Troyes, a city of about 60,000 inhabitants 100 miles east of Paris.

III. Empirical Strategy

Our evaluation of the ZFU2G program is based on an augmented DD framework. The outcome variables are several measures of establishment settlements, employment and wages which are all computed from exhaustive administrative datasets. In this section, we describe the data sources that we use, the unit of observation, the sample of treatment and control groups, and our econometric specifications.

A. *Variables of Interest*

In order to measure economic activity, we look at firm dynamics—through the number of establishment inflows, and at worker dynamics—through employment and wage numbers. Establishment settlements are measured with the administrative database SIRENE, managed by the French National Institute of Economics and Statistics (hereafter, INSEE), which gives information about the stock of establishments every first day of the year, and about the flow of new settlements during the year. An establishment is a legal entity, with no limit of size, which is precisely defined as being registered in SIRENE. We restrict ourselves to private-sector establishments, which are the only target of the ZFU program. In addition to a measure of total establishment inflows, it is possible to distinguish between pure "creations" or mere geographical "transfers" from elsewhere.[18]

Information on employment and wages is taken from another administrative database: the "Déclarations Annuelles de Données Sociales" (hereafter, DADS). The DADS come from the exhaustive collection of mandatory employer reports of the

[18] INSEE defines a "creation" as either the implementation of new production capacities (new establishment regimental number in SIRENE), or an establishment reactivation after a year without activity, or the resumption of incumbent capacities as long as at least two out of the three following establishment's features have changed: legal corporate entity, sector or location. By way of contrast, a "transfer" indicates a displacement of a preexisting establishment from one location to another.

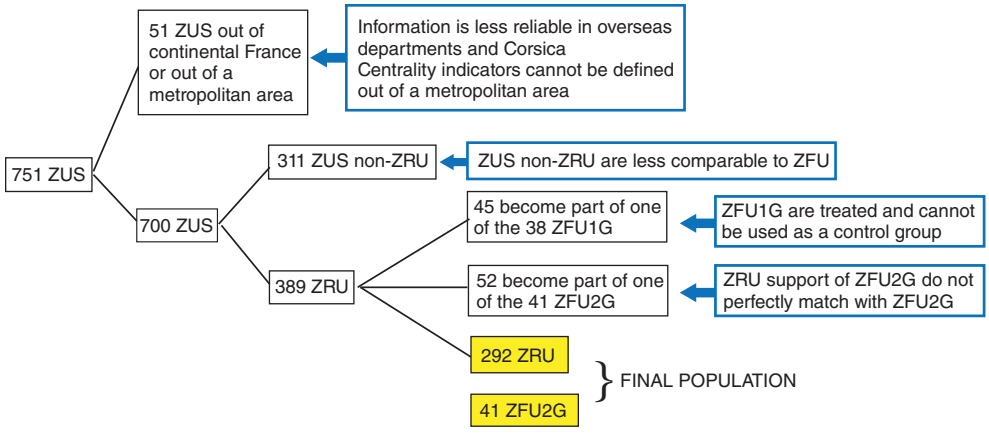


FIGURE 3. SELECTION PROCESS OF THE POPULATION OF NEIGHBORHOODS

earnings paid to each employee of the private sector subject to French payroll taxes. From this individual-level dataset, we distinguish between the extensive (number of jobs) and the intensive (numbers of hours worked) margins of employment.[19] We complement this analysis by studying the impact of the program on local earnings, computed as either the average or different percentiles of the hourly net wage over all establishments with registered employees located in the neighborhoods.[20] We attempt to address the mailbox effect by selecting establishments which are described as being both administratively and economically active for a given year. We also exclude seasonal activities.

B. *Sample*

Our selection process of the population of neighborhoods is described in Figure 3: starting from the 751 neighborhoods targeted as ZUS in 1996, we consider the 700 which are located in mainland France and are part of a metropolitan area. Out of those 700 ZUS, only 389 are also designated ZRU: we drop the 311 ZUS non-ZRU in order to keep the control group as homogeneous as possible. Among those ZRU, we drop the 45 which become part of a ZFU1G because they are treated by a previous wave of the program and therefore do not qualify as plausible control units. We also drop the 52 neighborhoods that will become part of the 41 ZFU2G and we replace this latter group with the 41 observations that correspond exactly to the ZFU2G. We need this step because ZRU and ZFU2G are not perfectly nested, even though they can be ranked from the viewpoint of policy intervention: ZFU were

[19] A job refers to the administrative notion of "post." One post corresponds to the role of one employee in a company. An employee who works in two different companies corresponds to two posts. This employee is thus counted twice. We only take nonannexed posts into account. A post is considered as nonannexed if the volume of work and the corresponding level of pay are deemed "sufficient." In most cases, if the pay is greater than three months of the minimum wage, or if the amount of time worked is greater than 30 days or 120 hours and the ratio of number of hours to total duration is greater than 1.5, the post is considered nonannexed.

[20] Between 30 and 40 percent of the total stock of establishments have at least one registered employee.

largely redrawn to be more inclusive than the original ZRU so as to include vacant land that could be used by new firms locating in these areas. Hence, ZFU2G are generally larger than the original ZRU from which they were selected. This leaves us with a set of 333 neighborhoods: 41 treated ZFU2G and 292 control ZRU. Note that this latter group includes the ZRU which will become part of the ZFU3G in 2006.

To perform a consistent statistical analysis at the neighborhood level, we use a smaller seed than the neighborhood, called "lots Regroupés pour l'Information Statistique" (hereafter, IRIS).[21] There are about 16,000 IRIS drawn within municipalities of more than 10,000 inhabitants, which is the minimum population threshold to be eligible for ZFU designation. To identify the relevant statistical information in the targeted neighborhoods and the control group, the boundaries of which do not match the IRIS partition, we choose a simple geographical allocation rule: any IRIS which intersects a ZFU "x" on more than 50 percent of its area will be considered as treated; similarly, any IRIS intersecting a ZRU "y" on more than 50 percent of its area will be considered as control.[22] We identify 231 treated IRIS and 932 control IRIS.[23] These 1163 IRIS form our estimation sample, unless there are missing values for certain outcome variables.

The number of years we include in our main estimation sample is largely driven by data limitations. Both databases SIRENE and DADS started being geocoded in 2002. However, whereas SIRENE also benefited from a retroactive geocoding for both firm stocks and flows for all the years following 1995, thanks to a strong public demand for information regarding local firm dynamics, this was not the case for the DADS stocks. Consequently, the precise location of each worker, obtained by the matching of DADS database with the database on stocks is only really observable after 2002. Before that, stocks of workers are only available every other year, which makes it impossible to compute yearly growth rates. For consistency, we also restrict the sample over this period for establishment inflows. It is, in theory, possible to study establishment inflows over a longer period (and actually, we do perform a robustness check on the pretreatment effects of the program, which is precisely based on this feature). However, this dataset extended into the past has to be taken with caution because the retroactive geocoding process is not exactly similar. In addition, there also are spatial and sectoral inconsistencies that jeopardize the study of establishment inflows at later dates than 2006. First, a few ZFU2G were enlarged in 2007 but this extension was not taken into account in the 2007 wave of SIRENE. Second, the nomenclature of the different sectors has also changed a lot in 2007. For all these reasons, we believe it is safer to restrict our empirical evaluation to the period 2002–2006.[24]

[21] This choice for the unit of observation is also the one made by the pioneer study of Rathelot and Sillard (2007), who argue that it has two advantages. First, the IRIS partition is rather homogenous regarding the size of the different units, which mitigates the risk of serious Modifiable Areal Unit Problem (Briant, Combes, and Lafourcade 2010). Second, these units are both large enough (around 2,000 inhabitants, on average) to limit the number of missing values and to reduce the level of spatial autocorrelation, and small enough to allow for a precise description of the zones under study.

[22] A sensitivity analysis of our results to the choice of this threshold is provided in an online Appendix.

[23] Figure A3 in Appendix A provides an example of the various spatial configurations between IRIS and ZFU.

[24] Descriptive statistics on the SIRENE and DADS data are provided in Appendix C.

C. *From Topographic Indicators to a Global Index of Spatial Isolation*

Each of our geographical indicators aims to describe one specific feature of spatial isolation. However, these indicators are correlated with each other in various ways and therefore, it is perilous to include several of them in a single regression. Moreover, we seek to have a general discussion about the three aforementioned dimensions of spatial isolation (severance, accessibility, and centrality), rather than to pursue a tedious case by case study. Hence, we proceed to the aggregation of our indicators into four general indices: one per dimension and one global index of isolation. Our aggregation strategy is based on the rank of each neighborhood in the distribution of the various isolation indicators within the set of 333 treated and control neighborhoods used in the econometric evaluation. This amounts to assuming that the relative position of each neighborhood in terms of geographical isolation matters more than the exact value of the indicator. In that sense, it is a standardization that transforms each indicator into a uniformly-distributed index except that, by convention, tied neighborhoods are assigned the same rank. Then, we simply aggregate these indices by taking their average value by dimension and subdimension.

We define a severance index as the average between a cutoff index and a buffer index. More precisely, let I_{cut} be the set of spatial indicators *id* described in panel A of Table 1. For each $id \in I_{cut}$, we compute the rank of the neighborhood within the distribution of all treated and control neighborhoods and then, the average value of these ranks: this yields a sub-index id_{cut} . We proceed the same way to obtain id_{buf} from the indicators in panel B (see Table 1). Finally, we compute the severance index as $id_{sev} = (id_{cut} + id_{buf})/2$. The accessibility index is defined along the same lines, as the average between a distance index and a catchment area index.[25] Finally, the centrality index is the average of the ranks associated with the four market potentials presented in Table 1. These three indices of spatial isolation will be used as such in the rest of the paper. However, in order to get an even more global sense of the isolation of each neighborhood, we also aggregate them into a global isolation index. We adjust the computation of the accessibility and centrality indices so that they work in the same direction as the severance index. Consequently, this global isolation index increases with the level of spatial isolation: the higher the index, the more isolated the neighborhood.[26] The distribution of the four indices is displayed in Figure 4.

By construction, all indices have the same mean (167). The global index of isolation is almost normally distributed. Its variance is reduced compared to the other three, which stems from the fact that our aggregation strategy does not take the correlation between the different dimensions into account. The centrality index is almost uniformly distributed, which means that the four market potential indicators used to construct this index are strongly positively correlated with each other. Finally, a noteworthy feature of the accessibility index is its double-humped shape,

[25] However, note that these two subdimensions (distance to infrastructures and catchment area) work in opposite directions: while the former indicates more isolation, the latter indicates less isolation. Therefore, we reverse the distributions of the indicators of distance, so that the index can be straightforwardly interpreted as a positive indicator of accessibility. Therefore, $id_{acc} = (id_{catch} + 334 - id_{dist})/2$.

[26] All datasets and program files are available online and the reader is invited to construct alternative indices and test their relevance in our empirical framework.

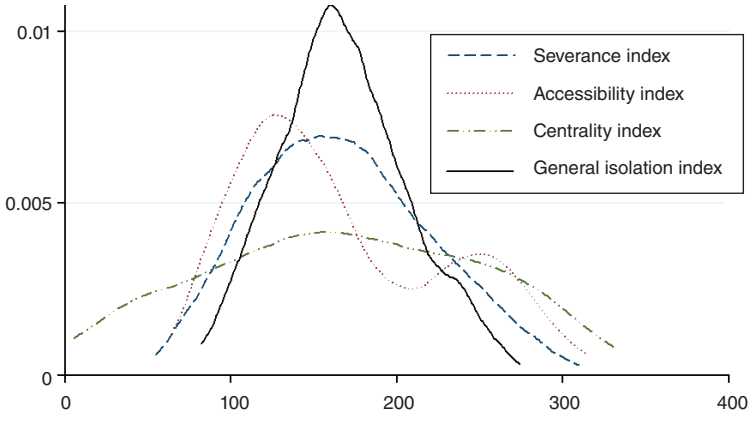


FIGURE 4. DISTRIBUTION OF THE FOUR INDICES OF SPATIAL ISOLATION

Note: The sample is made of 292 control ZRU and 41 treated ZFU2G.

Sources: GIS SG-CIV and BD-TOPO

which is due to the fact that many neighborhoods do not benefit from any train or metro station in their immediate surroundings: they are therefore identical with respect to the two indicators provided in panel D.[27] In the econometric evaluation, these four indices will be standardized according to their distribution within the population of treated IRIS.

D. *Econometric Specifications*

We assume a parsimonious econometric model where neighborhood characteristics only impact the level of the outcome variables, but not its evolution. As shown by Rathelot and Sillard (2007), this assumption is plausible if the outcome variables are in growth rates. Therefore, in our econometric study, we evaluate the impact of the ZFU2G program on the annual log differences of these outcomes. The specification we estimate is the following:

$$\Delta Y_{i\tau} = Y_{i\tau} - Y_{i\tau-1} = \alpha_\tau + \beta T_{i\tau} + \varepsilon_{i\tau}, \tag{2}$$

where i is an IRIS located in a neighborhood $z(i)$ (ZRU or ZFU2G), τ is the observation year, Y is the economic variable of interest in log, α_τ is a time dummy capturing conjuncture effects, and $\varepsilon_{i\tau}$ is the error term. The treatment variable $T_{i\tau} = 1_{\tau \geq t_0} \times 1_{i \in ZFU2G}$ is a dummy equal to 1 for every IRIS hosted by a ZFU2G observed after t_0, the implementation date of the program (2004). The coefficient β of this linear regression gives the average treatment effect under the assumption that

[27] To show that these somewhat abstract indices reflect real situations, two extensions are proposed in an online Appendix: first, we show that the topography of our two aforementioned polar examples of ZFU ("Les 4,000" and "Chantereigne-Montvilliers") is well captured by our indices; second, we show that these indices are correlated with individual mobility patterns across ZFU and ZRU.

TABLE 2—GEOGRAPHICAL DIFFERENCES BETWEEN ZFU2G AND ZRU

	ZFU2G	ZRU	Difference	Coeff. ZFU2G
Severance index	196	163	33**	13*
Accessibility index	187	164	23**	7
Centrality index	145	170	−25*	7
Global isolation index	177	165	12	−0.17

Notes: (i) Statistics based on the comparison between the 41 ZFU2G and the 292 control ZRU; (ii) Column "Difference" gives the significance of the difference between the first two columns; column "Coeff. ZFU2G" gives the coefficient associated with the dummy variable "ZFU2G" in an OLS regression of the spatial indicator described in line, on this dummy variable and MA fixed effects.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Sources: GIS SG-CIV and BD-TOPO

both treated and untreated units would have followed the same trend in the absence of treatment.[28]

In order to study whether geography can account for part of the heterogeneity in the treatment effect, we first consider the separate estimation of equation (2) on two subsamples of IRIS, selected according to their position in the distribution of the global isolation index $G_{z(i)}$ described in Section IIIC. Let $g(i) \in \{H, L\}$ denote the type of each IRIS, which can be spatially-integrated (H, if its global isolation index value is lower than the median value) or spatially-isolated (L, if its global isolation index value is larger than the median value). Our first specification involving geography is then:

$$\forall g(i) \in \{H, L\},\ \Delta Y_{i\tau} = \alpha_{g(i)\tau} + \beta_{g(i)} T_{i\tau} + \varepsilon_{i\tau}. \tag{3}$$

The comparison of the estimates of β_H and β_L is a first way of assessing the differential impact of geography on the effectiveness of the ZFU program.

However, this identification strategy raises a selection issue. Indeed, it may well be the case that some geographical characteristics are correlated with unobserved criteria of selection into the program and are therefore not perfectly orthogonal to the ZFU designation. Table 2 illustrates this issue. The first three columns, which display the results of an unconditional t-test of sample means between ZFU2G and ZRU, sometimes exhibit substantial differences, which cast doubt over the validity of our assumption. However, the fourth column shows that, conditional on both neighborhoods being located in the same MA, these differences tend to disappear.[29]

Our solution to this problem is to include neighborhood fixed effects to control for the potential impact of unobserved geographical heterogeneity on the evolution (and no longer only on the level) of our outcome variables. However, given the high

[28] In an online Appendix, we also look at the time-evolution of the treatment effect.

[29] A second issue pertains to the quality of the geographical information. Since the BD-TOPO is continuously updated without historical records for previous years, we were only able to use the 2006 version, which was the only version available at the time. Subsequently, it may be the case that some of the geographical features of neighborhoods, such as closeness to transportation infrastructures, have been partly caused by, or at least jointly determined with, the ZFU program. However, given the amount of time required to substantially modify the geography of a neighborhood, we believe this is of second order.

number of neighborhoods (333 in the full sample), we can only safely do so if we no longer split the estimation sample according to $g(i)$, so that we keep as many degrees of freedom as possible. To this end, we move towards a more parametric DD model à la Kolko and Neumark (2010), estimated on the whole sample of ZFU2G/ZRU, in which we interact the treatment variable T with a standardized version of $G_{z(i)}$. We also include fixed effects $\zeta_{z(i)}$ defined at the ZRU/ZFU2G level. This interacted model is summarized by equation (4):

$$\Delta Y_{i\tau} = \alpha_\tau + \zeta_{z(i)} + \beta T_{i\tau} + \lambda T_{i\tau} \times G_{z(i)} + \varepsilon_{i\tau}. \tag{4}$$

Given the possible correlation between our indices of spatial isolation, the analysis of the effect on one single index at a time could be justified. However, it may not be enough to account for the global complexity of the phenomenon under study. Therefore, we seek to study simultaneously the impact of urban severance, accessibility and centrality. For this reason, we consider an augmented model where the treatment variable $T_{i\tau}$ is now interacted with the three different indices described in Section IIIC, denoted $G^k_{z(i)}$. Depending on whether we allow for fixed effects in this model or not, our specification is either given by equation (5) or by equation (6).

$$\Delta Y_{i\tau} = \alpha_\tau + \beta T_{i\tau} + \sum_k \eta_k G^k_{z(i)} + \sum_k \lambda_k T_{i\tau} \times G^k_{z(i)} + \varepsilon_{i\tau}, \tag{5}$$

$$\Delta Y_{i\tau} = \alpha_\tau + \beta T_{i\tau} + \zeta_{z(i)} + \sum_k \lambda_k T_{i\tau} \times G^k_{z(i)} + \varepsilon_{i\tau}. \tag{6}$$

IV. Results

This section reports the estimation results of equations (2)–(6) regarding the impact of the program on the growth rate of the seven outcome variables described in Section IIIA: total inflow of establishments, creation of establishments, transfer of establishments, number of jobs, number of hours worked by the local workforce, average wage among the local workforce, and the ninetieth wage percentile of the local workforce. The growth rates of settlements, regardless of the establishment type, refer to the past year. For example, the outcome value for 2004 is the difference between the log of establishment inflows during 2003 and 2004. As it is computed for each year between 2003 and 2006, the impact of ZFU2G on settlements can be estimated for the first three years of treatment (i.e., 2004, 2005, and 2006). By way of contrast, jobs, hours, and wages are measured at the beginning of each year. Their values for 2004 are thus the difference between the logs of the related values on January 1, 2005 and January 1, 2004. For this reason, the impact of the ZFU2G program on jobs, hours, and wages can only be observed with a one-year delay in comparison with establishment inflows.

A. *Main Result*

Table 3 reports the DD estimates of the average impact of the program on establishment settlements, employment, and wages. Standard errors are clustered at the

Table 3—The Differential Impact of the ZFU2G Program According to Spatial Isolation

	Inflows (1)	Creations (2)	Transfers (3)	Jobs (4)	Hours (5)	Wage (6)	P90 (7)
Panel A: Average impact of the ZFU2G program							
Treatment	0.085**	0.033	0.184***	0.085***	0.050	0.0044	0.017**
	(0.041)	(0.036)	(0.052)	(0.030)	(0.034)	(0.0055)	(0.007)
Observations	3,301	3,256	1,379	2,714	2,714	2,714	2714
Number of MA	110	110	93	109	109	109	109
R^2	0.013	0.006	0.020	0.023	0.021	0.004	0.004
Panel B: Average impact in spatially-isolated neighborhoods							
Treatment	0.0071	−0.024	0.121**	0.060*	0.020	0.010*	0.023**
	(0.038)	(0.039)	(0.046)	(0.030)	(0.037)	(0.0058)	(0.010)
Observations	1,588	1,574	664	1,286	1,286	1,286	1286
Number of MA	48	48	41	48	48	48	48
R^2	0.005	0.003	0.014	0.020	0.019	0.005	0.01
Panel C: Average impact in spatially-integrated neighborhoods							
Treatment	0.163***	0.091***	0.245***	0.114***	0.085**	−0.0002	0.011
	(0.029)	(0.030)	(0.053)	(0.041)	(0.042)	(0.0092)	(0.012)
Observations	1,713	1,682	715	1,428	1,428	1,428	1428
Number of MA	83	83	69	82	82	82	82
R^2	0.026	0.013	0.030	0.027	0.026	0.003	0.002

Notes: (i) In parentheses: standard errors clustered at the metropolitan area level; (ii) Treatment stands for "average treatment"; Wage is mean hourly wage and P90 is the ninetieth percentile of hourly wage. The estimates correspond to the coefficient β in equation (2) panel A and to the coefficients $\beta_{g(i)}$ in equation (3) panels B and C; (iii) The sample sizes in panels B and C slightly differ because the IRIS associated with the median value of the global isolation index are all included in Panel C; results stay unchanged if these IRIS are assigned to panel B instead.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Sources: SIRENE and DADS, GIS SG-CIV and BD-TOPO

metropolitan area level to account for the issue of spatial correlation. Since the IRIS used in the estimation are located in more than at least 40 different metropolitan areas, the bootstrap procedure described in Busso, Gregory, and Kline (2013) is not necessary. To assess the role of geography for the success or failure of the ZFU2G program, we split the sample related to each of our seven outcome variables into two subgroups of more or less isolated set of neighborhoods according to the median of the global isolation index presented in Section IIIC. Panel A describes the result on the general population of neighborhoods, whereas panels B and C report the treatment estimated for the two subpopulations.

The average impact of the ZFU2G program is an additional 8.5 percent in the growth rate of establishment inflows in comparison with the control group during the period 2004–2006 (see panel A). This estimate is of the same order of magnitude as the 5 to 8 percentage point propensity-score difference found by Givord, Rathelot, and Sillard (2013). Although positive, this effect is around ten-fold lower than that found for ZFU1G Givord, Quantin, and Trevien (2012).[30] The growth

[30] Recall that this large difference may be due to the fact that, contrary to the evaluation of ZFU1G, both treatment and control groups have here already been treated by the ZRU program for eight years before the

rate of transfers is significantly higher in ZFU2G than in ZRU (+18.4 percentage points), whereas this is not the case for creations from scratch, even though they form the lion's share (around 80 percent) of total establishment inflows for any given year. The stronger impact on transfers is an indication that some firms have taken advantage of the program to displace preexisting establishments: the main effect of the program is therefore to foster the relocation of incumbent firms, and not to create new production capacities. This result is also perfectly in line with Givord, Rathelot, and Sillard (2013) and Mayer, Mayneris, and Py (2012), who both show that the ZFU2G program is very likely to have had negative spillovers on the economic performance of the neighboring areas.

The ZFU2G program had a stronger impact on job creation (+8.5 percentage points) than on hours worked (no significant impact). As already mentioned, most tax breaks in ZFU are targeted to new hires in the limits of 50 employees, but tax exemptions tend to decrease with the work task. The smaller impact found on the intensive margin of employment is therefore perfectly consistent with the design of the program. It is also consistent with the definition of jobs that we use, which may not be full-time jobs (see Section IIIA). If employers face convex costs with the number of hours worked per worker, they will tend to react to the policy by increasing the number of jobs, without necessarily increasing the total number of hours worked.[31] The lack of impact on average earnings confirms this finding. However, it is largely determined by the wage compression at the minimum wage in these areas in France. Column 7 shows that if, instead of using the growth rate of the mean wage, we use the growth rate of the ninetieth percentile, we find a significantly positive, but rather modest, impact of the program on the upper tail of the distribution of earnings, which are not bound by the minimum wage: ZFU2G have experienced an additional 1.7 percentage points in top-wages in comparison with ZRU.

The impact of the ZGU2G program is statistically different for the two sets of neighborhoods. Tax incentives are very effective at attracting new businesses into spatially-integrated neighborhoods: the growth rate of establishment inflows is 16 percentage points higher in ZFU2G than in ZRU, an order of magnitude that is twice the average treatment effect. More importantly, while the program fails to generate creations from scratch on average, it succeeds in fostering company births in the less isolated neighborhoods (+9.1 percent point difference in favor of treated areas). By way of contrast, only transfers contribute to the flow of incoming establishments in the more isolated neighborhoods, which clearly reduces the effectiveness of tax exemptions there. It is also worth noting that the program increases significantly the intensive margin of employment as long as the neighborhood is not too spatially isolated: the less isolated ZFU have benefited from an additional 11.4 percentage points in job creations, and an additional 8.5 percentage points in

implementation of the ZFU2G, and many firms may have already seized the opportunity. Moreover, given that ZRU keep benefiting from tax exemptions (even though far less generous than in ZFU) after 2004, the effect of the ZFU2G program is likely to be understated in our evaluation.

[31] Additional regressions on the job creation rate split between different skill levels, available upon request, show that the most-impacted category of workers, at least in terms of growth rates, is white-collars. Given the qualification mix of the worker population in firms located in these neighborhoods, this stronger impact on the more qualified workers is diluted by studying the evolution of total employment.

hours worked whereas there is no marked difference between the treated and control groups in the more isolated neighborhoods.

Wages stand as a critical exception.[32] As shown by panel B, the ZFU2G program has generated a 1 percent increase in the growth rate of average earnings of isolated communities, whereas the same outcome was not significantly affected in connected neighborhoods. The impact of the program on the growth rate of the ninetieth percentile is also very sensitive to geography, between a nonsignificant 1.1 percent estimate in spatially-integrated neighborhoods and a highly significant 2.3 percent estimate in the more isolated neighborhoods. The important conclusion we draw from this result is that isolated treated neighborhoods have more of a response in terms of earnings than in terms of jobs. As pointed out in Kline (2010), job variations are actually a measure of the distortions induced by EZ programs. But the benefits to individuals are better grasped by changes in earnings. Because fewer outsiders are willing to commute in more isolated neighborhoods, those areas reap benefit from this inelastic labor supply through an increase in wages. This result is very enlightening from a policy point of view, as it provides a clear rationale for still targeting isolated areas, despite the ineffectiveness of the program on many other aspects. Yet, this optimistic message must be taken with a certain degree of caution: the wage premium generated by the ZFU2G program for people working in isolated neighborhoods is rather low in comparison with the variation of other outcomes and it is mostly driven by the higher-paying jobs. Regardless of geography, the program does not appear to increase earnings for low-paid workers, even though they make up for a very large share of the workforce in these neighborhoods.[33]

B. *Robustness Checks*

The DD approach is relevant only if the treated and control groups share a common trend before the program is enacted. Graph A in Figure 5 gives the evolution of the mean of one of our variables of interest: the annual number of establishment inflows by neighborhood, both in the ZRU that became ZFU2G in 2004 and in the ZRU that were not treated. We take 1998 as the baseline year since the ZFU program was enacted in 1996 and our outcome variables are growth rates. By 2006, the annual inflow of establishments had been multiplied by more than two in ZFU2G and by almost 1.5 in ZRU. The trends of the two groups start to diverge in 2004, which corresponds to the first year of the ZFU2G treatment. We compute the 90 percent confidence intervals of this average value for ZFU2G and ZRU. By 2004, these two confidence intervals no longer intersect. Graph B gives support to our claim that the ZFU program had a very heterogeneous impact across locations. The solid lines depict the mean number of establishment inflows for the more integrated neighborhoods, as selected by the value of their isolation index $G_{z(i)}$ and the dashed lines report the same number for the more isolated neighborhoods. Whereas the levels and trends of the four groups cannot be distinguished prior to 2004, the

[32] We thank a referee for pinning down this point and encouraging us to investigate in this direction.

[33] In 2007, at least one out of four new hires in ZFU2G and ZRU was paid at the minimum wage and one out of two did not exceed a 10 percent cap of this value (Quantin 2012).

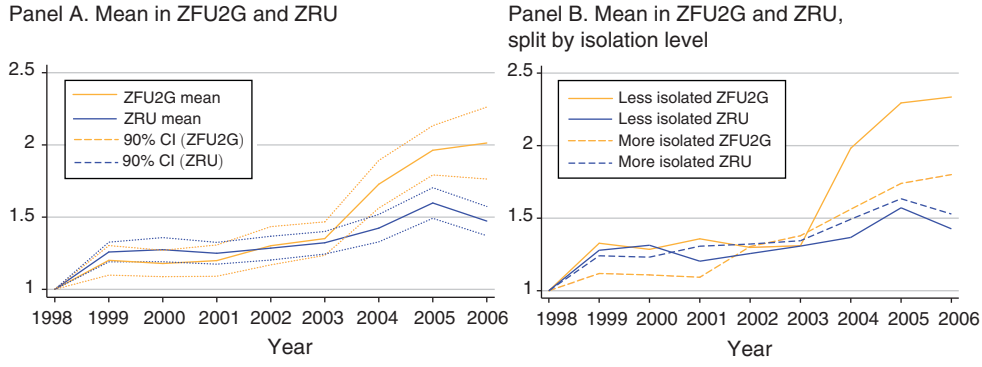


FIGURE 5. ANNUAL INFLOW OF ESTABLISHMENTS IN ZRU AND ZFU2G

Notes: (i) Each point is the ratio of the moment described for a given year to its counterpart in 1998; (ii) ZFU2G and ZRU here stand for the IRIS approximation of ZFU2G and ZRU; (iii) Graph 2 is based on the same partition as equation (3), and Table 3.

Source: SIRENE

group of integrated ZFU2G is the only one to experience a sharp increase in 2004. By 2006, annual inflows in this group have been multiplied by more than 2.3 with respect to 1998 whereas they have been multiplied by 1.5 in both groups of ZRU, and by less than 1.8 in the more isolated ZFU2G. Hence, the impact of the ZFU program on establishment inflows is almost entirely driven by the effect on integrated neighborhoods.

In addition to this graphical evidence, we perform a falsification test to see whether the ZFU2G policy had an impact during all the pretreatment years. Because of the data limitations described in Section IIIA, we are only able to do it for the three establishment variables. Table D1 in Appendix D reports the yearly impact of being located in a ZFU2G prior to the implementation of the ZFU2G program. Even though we have to remain cautious in the way we interpret these figures due to data inconsistency over this long-period (the test is valid only if the noise created by changing the geocoding process is not correlated with the localization of firms), this table shows that the estimated coefficients are not significant for 1998 to 2003, regardless of the outcome studied, which gives additional support to our identification strategy.

Admittedly, one might also argue that the magnitude of our estimates depends on the control group. As a further robustness check, we provide the difference in outcomes between the treatment group and the subset of ZRU lately designated ZFU3G, that might share common unobserved characteristics with ZFU2G. As shown in panel A of Table 4, results remain mostly unchanged: in comparison with ZFU3G-to-be, ZFU2G experienced a 10.9 percent point increase in establishment settlements, that were mainly driven by transfers. However, with this new control group, the program had no visible effect on job creation or high-earnings.

Another issue could be that using nearby ZRU as controls exaggerate the positive impact of the program, because of negative spatial spillovers due to the "cannibalization" of surrounding activities. If the main effect of the ZFU2G program is

TABLE 4—ROBUSTNESS CHECKS ON CONTROL GROUPS

	Inflows (1)	Creations (2)	Transfers (3)	Jobs (4)	Hours (5)	Wage (6)	P90 (7)
Panel A: Impact on all ZFU2G, control by ZFU3G							
Treatment	0.109*	0.031	0.170**	0.016	-0.033	0.0008	0.008
	(0.053)	(0.052)	(0.072)	(0.054)	(0.065)	(0.010)	(0.019)
Observations	1,393	1,380	641	1,131	1,131	1,131	1,131
Number of MA	35	35	34	35	35	35	35
R^2	0.019	0.008	0.027	0.026	0.028	0.007	0.007
Panel B: Impact on ZFU2G, control by ZRU in MA without ZFU2G							
Treatment	0.086**	0.021	0.235***	0.090***	0.054	-0.0029	0.011
	(0.039)	(0.036)	(0.050)	(0.033)	(0.037)	(0.0063)	(0.0095)
Observations	2,314	2,284	980	1,882	1,882	1,882	1,882
Number of MA	110	110	93	109	109	109	109
R^2	0.019	0.009	0.027	0.029	0.027	0.004	0.006
Panel C: Impact on SW and NE ZFU2G, control by NW and SE ZRU							
Treatment	0.083*	0.043	0.137***	0.095**	0.059	0.0069	0.020
	(0.042)	(0.040)	(0.050)	(0.042)	(0.055)	(0.0091)	(0.013)
Observations	1,505	1,485	598	1,228	1,228	1,228	1,228
Number of MA	62	62	51	60	60	60	60
R^2	0.011	0.009	0.022	0.017	0.018	0.006	0.006
Panel D: Impact on NW and SE ZFU2G, control by SW and NE ZRU							
Treatment	0.091*	0.029	0.218***	0.081*	0.040	0.0025	0.017
	(0.046)	(0.040)	(0.066)	(0.041)	(0.040)	(0.0082)	(0.012)
Observations	1,796	1,771	781	1,486	1,486	1,486	1,486
Number of MA	67	67	59	67	67	67	67
R^2	0.014	0.005	0.029	0.030	0.025	0.004	0.005

Notes: (i) In parentheses: standard errors clustered at the metropolitan area level; (ii) Treatment stands for "average treatment"; Wage is mean hourly wage and P90 is the ninetieth percentile of hourly wage. The estimates correspond to the coefficient β in equation (2).

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Sources: SIRENE and DADS

to divert firms located nearby the ZFU border, our DD estimates do not take into account the welfare reduction possibly associated to these relocations for depleted ZRU. To tackle this issue, we run three further robustness checks. The first is to compare ZFU2G to ZRU located in metropolitan areas that do not host any ZFU2G. Panel B of Table 4 reports the related estimates. The significance and magnitude of the treatment are very similar to those reported in Table 3.[34] There is a potential caveat with this new control group however, since the average impact of the program might be driven by the cannibalization occurring within specific regions. To overcome this issue, we also compare ZFU2G to distant ZRU. To that end, we consider a partition of France in four quadrants defined by geographical coordinates and we

[34] If the coefficient on the growth rate of P90 is no longer significant because of the smaller sample size, the estimated values on the two subsamples are very close to those found in Table 3.

only compare ZFU2G to ZRU located in another quadrant.[35] Panel C in Table 4 reports the impact estimates of ZFU2G located in the southwest and northeast quadrants compared to the ZRU located in the southeast and northwest quadrants, whereas panel D considers the reverse population. The treatment estimates drawn from restricting the control group to distant ZRU are very close to those found with a less restrictive control group.

The problem with this solution is that it forces us to compare neighborhoods that are potentially more different from each other. Because of this tradeoff between avoiding contamination due to spillovers and comparability between treatment and control groups, we perform an additional robustness check, based on the tightening or loosening of the selection rule used to define treated and control neighborhoods, up to more than 90 percent of the IRIS overlapping the ZFU2G/ZRU. Results are reported in an online Appendix. Choosing a more stringent selection criterion reduces the probability to use nearby ZRU as controls. However, as soon as the selection criterion is above 10 percent of intersection, the average treatment effect is virtually unaffected. All of these robustness checks strengthen the conclusion that the overall impact of the ZFU2G program on our different outcomes, as displayed in the panel A of Table 3, is precisely identified and not overestimated.

Finally, we test the robustness of the findings of panels B and C in Table 3 to the inclusion of ZRU/ZFU2G fixed effects as described in equation (4). Table D2 in Appendix D reports the related estimates. The estimates are in line with the average impact depicted in Table 3, despite an increase in the standard errors due to the number of constraints imposed by a fixed-effects specification. The only exception comes from the wage equations, where the interaction terms fade away.

C. *Additional Heterogeneity*

The policy analysis can be refined by investigating how different dimensions of spatial isolation interact with the program effect for sectors with different requirements regarding the mobility and intensity of capital or labor.

The Different Dimensions of Spatial Isolation.—In order to deepen our understanding of the factors driving the impact of spatial isolation, we conduct a separate analysis for each of the three dimensions of spatial isolation presented in Section II: severance, accessibility, and centrality. Table 5 depicts the results of this multivariate estimation, based on equation (5). In order to ease the interpretation of the coefficient associated with the interaction of treatment and geography, indices of spatial isolation have been standardized within each estimation sample according to their distribution among IRIS located in ZFU2G.

Severance inhibits the effectiveness of the ZFU program to attract firms in the treated neighborhoods. The communities that are more isolated from the rest of the metropolitan area by cutoffs of any kind clearly underperform: for instance, a one standard deviation increase in the severance index reduces by 5.7 percentage

[35] South encapsulates all geographic coordinates with a latitude below $48.53^{o}N$, whereas West is defined according to a longitude above $2.50^{o}W$.

Table 5—The Role of Severance, Accessibility and Centrality

	Inflows (1)	Creations (2)	Transfers (3)	Jobs (4)	Hours (5)	Wage (6)	P90 (7)
Treatment	0.080***	0.032*	0.183***	0.081***	0.052*	0.0076	0.019***
	(0.014)	(0.017)	(0.032)	(0.024)	(0.027)	(0.0046)	(0.007)
Severance	0.0062	0.0032	0.0065	0.00072	−0.0039	−0.0028*	−0.0002
	(0.0075)	(0.0077)	(0.016)	(0.0091)	(0.012)	(0.0017)	(0.003)
Treat. × Sev.	−0.057***	−0.047**	−0.087***	−0.016	−0.0090	−0.0089**	−0.021***
	(0.016)	(0.019)	(0.028)	(0.026)	(0.030)	(0.0044)	(0.006)
Accessibility	−0.011	−0.012	0.0068	0.0042	−0.0065	−0.0027*	−0.004
	(0.0064)	(0.0083)	(0.014)	(0.0067)	(0.0074)	(0.0015)	(0.003)
Treat. × Acc.	0.056***	0.062***	0.041	0.0097	0.021	0.002	0.006
	(0.013)	(0.018)	(0.033)	(0.021)	(0.022)	(0.0038)	(0.005)
Centrality	−0.026***	−0.017**	0.019	−0.0079	−0.0059	0.0051***	0.003
	(0.0072)	(0.0076)	(0.015)	(0.012)	(0.013)	(0.0017)	(0.003)
Treat. × Centr.	0.066***	0.051**	−0.045	0.024	0.039	−0.015***	−0.022***
	(0.016)	(0.020)	(0.033)	(0.033)	(0.034)	(0.0048)	(0.007)
Observations	3,301	3,256	1,379	2,714	2,714	2,714	2,714
Number of MA	110	110	93	109	109	109	109
R^2	0.018	0.010	0.023	0.024	0.023	0.008	0.006

Notes: (i) In parentheses: standard errors clustered at the metropolitan area level; (ii) Treatment stands for "average treatment"; Wage is mean hourly wage and P90 is the ninetieth percentile of hourly wage. The estimates correspond to the coefficients β, η_k, and λ_k in equation (5).

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Sources: SIRENE, DADS, BD-TOPO, and GIS SG-CIV

points the growth rate of settlements in ZFU2G relatively to ZRU. Compared to the 8 percent point raise induced by the ZFU2G program on average, this reduction amounts to a program undereffectiveness of 70 percent.[36] As for accessibility, ZFU2G which are well-connected to the transportation network benefit from an additional 5.6 percentage points in the growth rate of establishment inflows. This effect is mainly driven by the entry of new business companies.[37] Public transportation services are thus likely to explain the relative success of the ZFU program in large metropolitan areas such as Paris, which benefit from good railroad passenger networks. Centrality also helps recover additional gains from tax breaks: the higher the market potential of a neighborhood, the more effective the ZFU program to

[36] It is difficult to draw from this kind of estimate a clear indication of how penalizing are urban enclaves. As a meaningful guidance for policymakers, we find it useful to estimate the penalty associated to each indicator used to construct the synthetic severance index. In a previous version of this work, we have shown for instance that an increase by one standard deviation in the number of impassable roads separating the ZFU border from the main CBD of the metropolitan area would translate into a 6.3 percent point decrease in the establishment settlement growth rate (Briant, Lafourcade, and Schmutz 2012).

[37] To take a more concrete example, it might be enlightening to know that a one-standard deviation increase in the number of train or metro stations within 500 meters of a ZFU, which is equivalent to adding two stations in the area, would be associated with a 5.8 percent point rise in the growth rate of establishment inflows.

generate new establishments from scratch. Table D3 in Appendix D shows that most of these results are robust to, if not strengthened by, the inclusion of neighborhood fixed effects.

Interestingly, by disentangling the effectiveness of the ZFU program between the three dimensions of spatial isolation, we get deeper into the mechanisms underlying its positive impact on the earnings of isolated neighborhoods. Whereas the interaction between the treatment and the centrality index exhibits a negative and highly significant impact on wages, which is perfectly in line with the aforementioned inability of outsiders to commute into peripheral neighborhoods, severance reduces the effectiveness of the treatment: the trade-off between firm/job creation in spatially-integrated neighborhoods versus wage stimulus in spatially-isolated neighborhoods is not relevant in the case of urban enclaves, where the ZFU program has little impact whatsoever.

The Interplay between Sectors and Geography.—The main finding so far is that urban geography affects the ZFU2G program in a less effective way regarding employment than regarding establishment settlements, just as the program is, on average, less effective on employment than on establishment settlements. However, this result hides a large heterogeneity between sectors. Previous works, such as Freedman (2013), have put the emphasis on the fact that the somehow modest positive effect of EZ programs was triggered by the low and middle-paying jobs created in the goods-producing, retail, and wholesale trade industries. Theory predicts that EZ programs should have less impact on more capital-intensive sectors on average, because firms in these sectors face higher sunk production costs and relocation costs. In addition, firms in more labor-intensive sectors should be more sensitive to spatial isolation. To investigate further this issue, we split our sample into different subgroups of industries. We use the two-digit industrial classification NAF17 provided by the INSEE, but keep out of the analysis those industries which exhibit a too small number of observations to provide robust estimates.

Table 6 breaks out the estimated impact of the ZFU2G program on the intensive margin of employment, by the most representative industries located in distressed neighborhoods. Panel A shows that the only statistically significant positive effect of the program is for the health sector (+21 percent in the growth rate of the number of hours worked). There is no discernible effect on the other industries. The impact of EZ programs on this sector has already been noted by policymakers, and it is also mentioned in Freedman (2013) and Mayer, Mayneris, and Py (2012). The health sector mostly encompasses self-employed workers, such as nurses or physicians. It is neither capital-intensive, nor space-consuming, unlike the manufacturing sector, which has larger establishments. As such, health firms are more sensitive to tax incentives and able to relocate more quickly, at lower cost.

In addition to its relatively high labor-intensity, other features may also explain why the medical sector is more impacted by the ZFU program: in particular, the relationship of health-related firms to their surroundings and the level of mobility in the daily activity of these firms. The French private health sector is characterized by a high level of mobility: for instance, self-employed nurses mostly work at their patients' homes, which are often located outside the ZFU. Whereas health and

Table 6—Sectoral Analysis: Average Impact of the ZFU2G Program on Hours Worked

	Manuf. (1)	Constr. (2)	Trade (3)	Hotels (4)	Transp. (5)	Health (6)
Panel A. Average impact						
Treatment	0.027	0.011	−0.014	−0.0052	0.012	0.213***
	(0.048)	(0.053)	(0.052)	(0.054)	(0.097)	(0.069)
Observations	1,598	2,054	2,319	1,308	961	1,415
Number of MA	99	104	109	93	81	91
R^2	0.002	0.009	0.001	0.007	0.005	0.023
Panel B. Average impact in spatially-isolated neighborhoods						
Treatment	−0.0090	0.0081	−0.074	0.067	−0.079	0.178*
	(0.061)	(0.050)	(0.068)	(0.096)	(0.087)	(0.10)
Observations	799	1,025	1,129	651	478	706
Number of MA	48	48	48	41	33	43
R^2	0.005	0.010	0.004	0.003	0.008	0.025
Panel C. Average impact in spatially-integrated neighborhoods						
Treatment	0.065	0.050	0.049	−0.089	0.111	0.236***
	(0.074)	(0.096)	(0.030)	(0.082)	(0.15)	(0.083)
Observations	799	1,029	1,190	657	483	709
Number of MA	72	77	82	69	60	69
R^2	0.007	0.026	0.002	0.021	0.006	0.024
Panel D. Impact of isolation with neighborhood fixed effects						
Treatment	−0.011	−0.080	−0.028	0.024	0.103	0.187**
	(0.073)	(0.076)	(0.043)	(0.10)	(0.13)	(0.087)
Treatment × Index	−0.056	−0.107**	−0.082***	−0.073	−0.073	−0.090*
	(0.052)	(0.049)	(0.023)	(0.056)	(0.10)	(0.053)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,598	2,054	2,319	1,308	961	1,415
Number of MA	99	104	109	93	81	91
R^2	0.117	0.086	0.090	0.104	0.213	0.120

Notes: (i) In parentheses: standard errors clustered at the metropolitan area level (panels A, B, and C); (ii) Treatment stands for "average treatment"; The estimates correspond to the coefficients β in equation (2) (panel A), to the coefficients $\beta_{g(i)}$ in equation (3) (panels B and C), and to the coefficients β and λ in equation (4) (panel D); (iii) Manuf.: manufacturing; Constr.: construction; Trade: Trade, retail and car repair shops; Hotels: Hotels and restaurants; Transp.: Transportation and communication; Health: Healthcare and social work.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Sources: DADS, BD-TOPO and GIS SG-CIV

social workers need to be able to commute into and out of the ZFU, it may not be as important to the people who, for instance, work at a manufacturing plant. Panels B to D show that, indeed, activity in the health sector is more likely to be boosted by the program in more integrated neighborhood. When we control for neighborhood fixed effects, additional patterns emerge regarding the construction and retail industries, which also both heavily rely on mobility: mobility of the workforce for construction (because construction often takes place in other parts of the city), and mobility of the customer base into the neighborhood for retail. In both cases, spatial isolation acts as a deterrent of the effectiveness of the program (panel D).

V. Conclusion

This paper attempts to give empirical support to the simple statement that local geography matters to the success or failure of place-based programs. Using a new dataset which describes the level of spatial isolation of the neighborhoods that were targeted by the French EZ program, it shows that this program was more likely to create jobs when the targeted neighborhoods were more spatially integrated and more likely to increase earnings when the targeted neighborhoods were more spatially isolated. If public authorities care first and foremost about boosting employment in these areas, they ought to target the least isolated among them. However, besides the political controversies this selection would raise, this choice may not be the most efficient. Given the crucial role played by spatial isolation, one can but acknowledge the necessity of combining place-based tax breaks and employment incentives with public investments in transportation infrastructure and services. The current "Grand Paris" global project, which aims at building a metro ring around the inner suburbs of Paris, as well as supporting sustainable economic and urban development, may be a right step in this direction. However, the cost of such investments (around 35 billion euros for the "Grand Paris" metro station project) calls for a cautious cost-benefit analysis that is difficult to undertake given that transportation infrastructures have multiple purposes other than reviving depleted urban communities.

Appendix A: Maps

A. *The Location of ZFU*

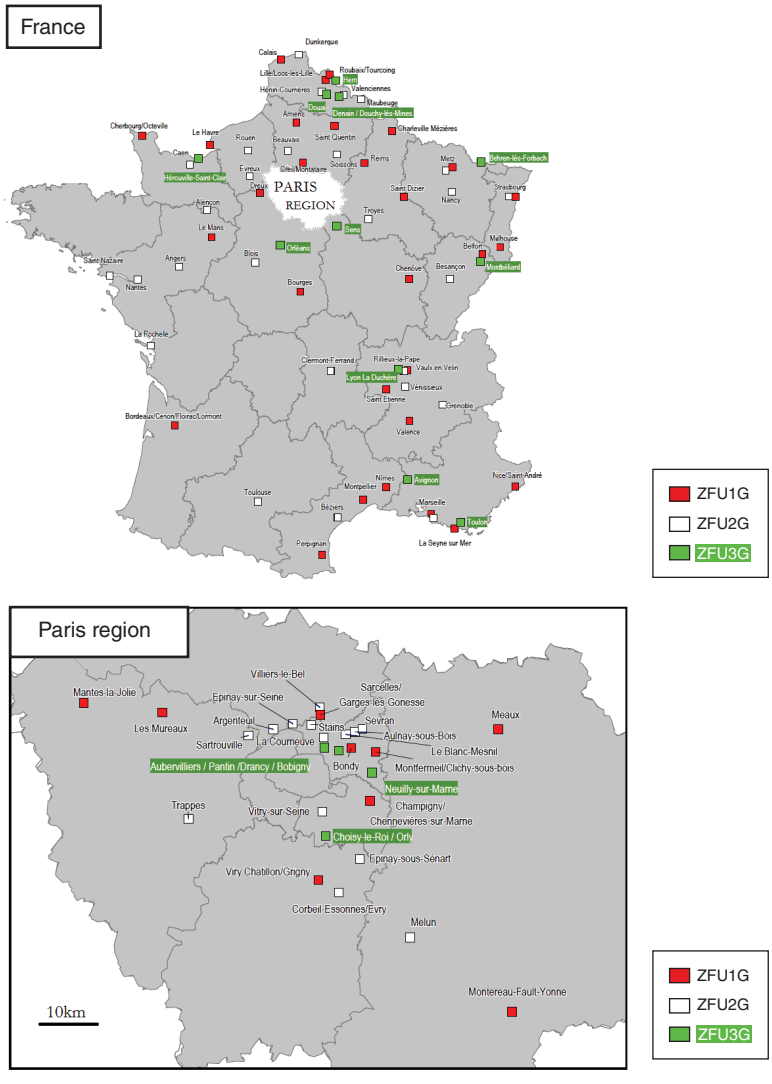


Figure A1. The 93 ZFU in Mainland France

Note: The names are the municipalities hosting a ZFU.

Source: GIS SG-CIV

B. *The Unit of Observation*

The ZFU "les 4,000" which is located northeast from Paris (see Figure A2), intersects with nine different IRIS. About half of the ZFU boundary match with IRIS boundaries. Six IRIS are more than 50 percent included in the ZFU (among which,

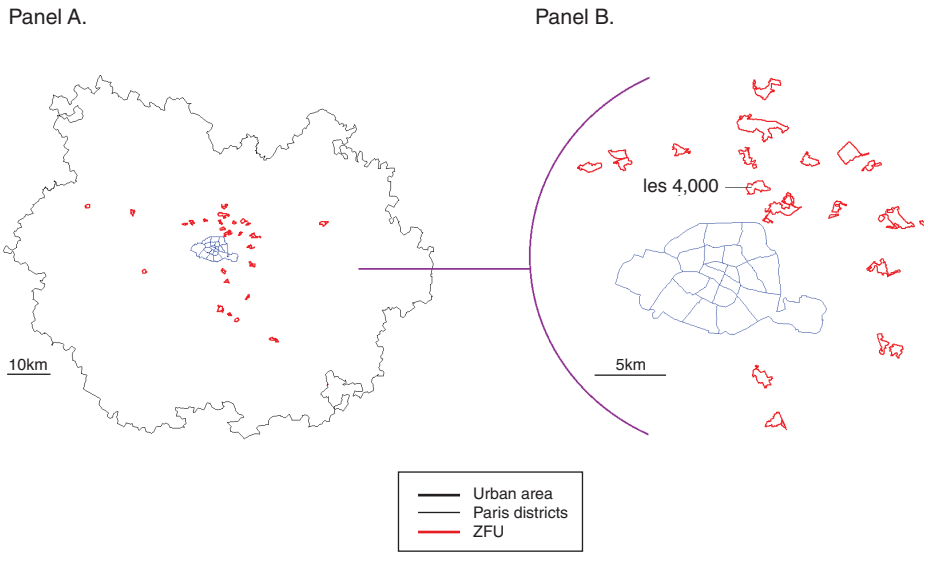


FIGURE A2. LOCATION AND SHAPE OF THE ZFU IN THE PARIS REGION

Source: GIS SG-CIV

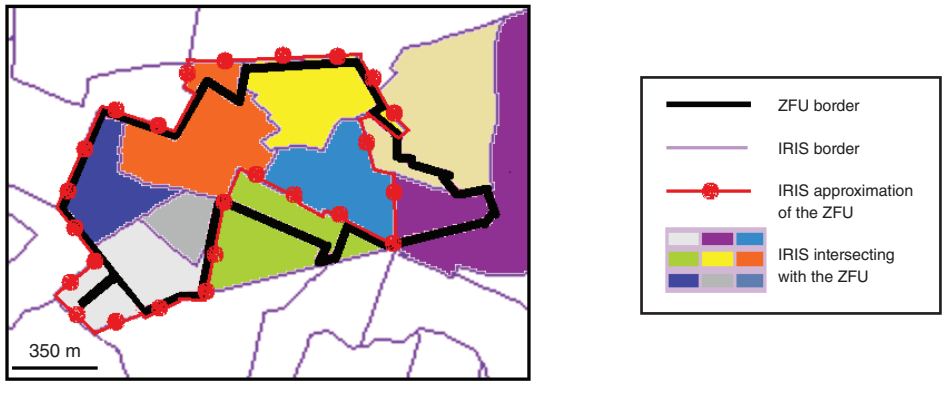


FIGURE A3. ZFU AND IRIS: THE EXAMPLE OF THE NEIGHBORHOOD "LES 4,000"

Source: GIS SG-CIV

three are entirely located within the ZFU), and they define the IRIS approximation of this ZFU.

APPENDIX B: THE TAX PACKAGE IN ZFU AND ZRU

To benefit from the tax package, firms have to declare, in their registrant's fiscal return, that they want to benefit from place-based exemptions. The relief is exclusive of any other existing exemption on low wages.

Payroll Taxes.—In ZFU, only firms with a turnover of less than 10 million euros, active plants preexisting or newly created in ZFU and less than 50 employees per ZFU are qualified to payroll tax exemptions. The relief concerns firms with less than 50 employees per ZFU in full-time equivalent, and it is limited to a cumulated 100,000 euros over three years. The deduction applies to *all employees enrolled in ZFU* with open-ended contracts or fixed-term contracts of more than 12 months. Those workers are exempt from employer payroll taxes only up to a cap of 40 percent above the legal minimum wage. This deduction applies to all ZFU employees for five years after the designation of the area as a ZFU or the settlement of a new establishment in the targeted area, and to all new ZFU recruits afterwards. In ZRU, there is no staff or turnover limitation for the firm, but the exemption only relates to *new workers recruited* in the establishment located in ZRU (and not to all workers, contrary to ZFU), it only lasts for 12 months and there is a maximum of 50 jobs in full-time equivalent.

Corporate Income Tax.—In ZFU, firms with a turnover of less than 10 million euros, active plants preexisting or newly created in ZFU and less than 50 employees per ZFU are eligible for corporate tax relief.[38] These firms are fully exempted for five years, within the limit of 100,000 euros per year (increased by 5,000 euros per new recruit). They are then exempted at a decreasing rate for the next nine years. In ZRU, tax relief is restricted to firms creating a new active plant in the area, with no staff limitations. These firms are fully exempted for two years, and exempted at a decreasing rate for the next three years. The total deduction is limited to 225,000 euros per 36-month period.

Local Business Tax.—In ZFU, only firms with a turnover of less than 10 million euros, active plants preexisting or newly created in ZFU and less than 50 employees per ZFU are eligible for business tax exemptions. For these plants, a full exemption applies for five years, within the limit of a taxable amount of 337,713 euros (2006 value) and 100,000 euros of cumulated aid over three years. They are then exempted at a decreasing rate for the next three to nine years, depending on the size of the firm (firms with less than five employees are eligible up to nine years). In ZRU, all businesses (with less than 150 employees) are eligible for five years. However, the ceiling of the taxable amount is lower (125,197 euros in 2006).[39]

Miscellanea.—All existing buildings located in ZFU belonging to firms liable for a property tax on buildings are exempt for five years. They are then exempted

[38] There are also sectoral restrictions, which for instance exclude the banking, finance, insurance, housing and sea-fishing sectors; however, these are minor restrictions given the sectoral mix in these neighborhoods (see Section IVC and Section C in the same Appendix for details).

[39] This disposition matters to the period under study but it is no longer relevant today. Until 2010, the local business tax ("Taxe professionnelle") was due each year by legal persons or self-employed people. It was levied to fund public services supplied by municipalities. The tax base was the rental value of the taxpayer assets, including the rateable value of its capital, equipment, and buildings. Local authorities, within limits set by the French national legislation, fixed the tax rate. In 2010, the business tax was replaced by a local economic contribution levied on the rateable value of property and value-added, to avoid hampering firms' investment (in the old system, the more a company invested, the larger its tax burden).

at a decreasing rate for the next three to nine years, depending on the size of the firm. No exemptions apply in ZRU. Firms in both types of neighborhoods may also be exempt from individual payroll taxes applying to self-employed craftsmen and salesmen (for up to five years). A minor addition in ZFU is that transaction costs related to the purchasing of a business or a clientele (for medical professions, for instance) are also reduced.

Appendix C: Descriptive Statistics

Table C1—The Economic Activity in ZRU/ZFU2G

Year	2003		2005	
Neighborhood	ZFU2G	ZRU	ZFU2G	ZRU
Panel A. Establishment inflows				
Total number	2,104	3,375	2,954	3,888
Share of creations (percent)	84.7	84.6	75.5	83.9
Manufacturing (percent)	4.3	5.0	4.3	3.8
Construction (percent)	27.1	25.2	26.9	26.1
Trade and retail (percent)	30.1	30.4	26.1	32.4
Hotels and restaurants (percent)	3.9	5.3	2.8	5.5
Transportation (percent)	6.4	5.9	5.3	5.7
Healthcare (percent)	5.8	7.2	9.9	6.1
Panel B. Establishment stocks				
Total number	9,611	17,187	10,879	18,117
Share with at least one employee (percent)	37.8	39.6	37.7	39.0
Manufacturing (percent)	6.2	7.0	5.6	6.3
Construction (percent)	15.4	15.4	16.7	15.6
Trade and retail (percent)	28.0	28.7	27.1	29.3
Hotels and restaurants (percent)	5.3	5.9	5.1	5.9
Transportation (percent)	7.6	6.7	7.5	6.8
Healthcare (percent)	17.1	15.8	16.4	15.3
Panel C. Employees				
Average number by establishment	12.7	12.4	11.4	11.0
White-collars (percent)	10.0	11.2	9.6	11.1
Intermediate professions (percent)	22.6	20.5	23.1	20.7
Blue-collars (percent)	67.4	68.3	67.3	68.2
Mean hourly net wage (€)	10.1	9.8	10.5	10.1
Median hourly net wage (€)	9.4	9.1	9.7	9.4

Notes: (i) ZFU2G and ZRU here stand for the IRIS approximation of ZFU2G and ZRU; (ii) stocks are computed on the first day of the year; (iii) the average number of employees by establishment is computed on the population of establishments with at least one employee.

Sources: SIRENE and DADS

Table C1 allows for a comparison between treated and control neighborhoods before and after the inception of the ZFU2G program. By 2005, there were about 11,000 establishments in the IRIS hold by ZFU2G. The sectoral mix showed the overrepresentation of the transportation, healthcare and construction industries, three sectors which require that a large part of the activity be conducted outside the neighborhood. With about 124,000 employees registered as working in the ZFU2G neighborhoods, this corresponds to an average establishment size of 11.4 employees —although a large share (around 60 percent) of the population of establishments

is actually composed of self-employed individuals. Most jobs located in ZFU2G require a low level of qualification, as depicted by the huge share of blue collars working in those areas (around 68 percent). Moreover, the wage distribution is very similar in ZRU and ZFU2G, but relatively compressed around the minimum wage in comparison with all the other neighborhoods that do not benefit from the Urban Stimulus Package.

Appendix D: Additional Regression Results

A. *Testing the Common Trend Assumption*

Table D1—Year-by-Year Falsification Test on Establishment Inflows

	Inflows (1)	Creations (2)	Transfers (3)
Pretreatment in 1998	0.011	0.056	0.038
	(0.049)	(0.048)	(0.096)
Pretreatment in 1999	−0.051	−0.021	0.045
	(0.054)	(0.047)	(0.11)
Pretreatment in 2000	0.019	−0.0025	0.00076
	(0.062)	(0.065)	(0.076)
Pretreatment in 2001	0.00017	−0.00086	−0.080
	(0.070)	(0.059)	(0.10)
Pretreatment in 2002	0.052	0.043	0.14
	(0.052)	(0.047)	(0.096)
Pretreatment in 2003	0.031	0.059	0.13
	(0.052)	(0.051)	(0.11)
Treatment in 2004–2006	0.085**	0.033	0.18***
	(0.041)	(0.036)	(0.052)
Observations	5,914	5,812	2,465
Number of MA	110	110	102
R^2	0.009	0.006	0.016

Notes: (i) In parentheses: standard errors clustered at the metropolitan area level; (ii) "Pretreatment in *X*" is a dummy variable equal to one if the IRIS is observed in year *X* and will become part of a ZFU2G in 2004; "Treatment in 2004–2006" is a dummy variable equal to one if the IRIS is observed either in 2004, 2005, or 2006 and is part of a ZFU2G. The estimates correspond to the coefficient β in equation (2).

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Source: SIRENE

B. *Robustness to the Inclusion of Neighborhood Fixed Effects*

Table D2—The Global Isolation Index and Unobserved Geographic Heterogeneity

	Inflows (1)	Creations (2)	Transfers (3)	Jobs (4)	Hours (5)	Wage (6)	P90 (7)
Treatment	0.044	−0.021	0.020	0.054*	0.029	0.016*	0.025*
	(0.033)	(0.038)	(0.091)	(0.032)	(0.034)	(0.0088)	(0.014)
Treat. × Index	−0.160***	−0.132***	−0.0041	−0.064***	−0.075***	0.0011	−0.003
	(0.011)	(0.016)	(0.023)	(0.021)	(0.021)	(0.0038)	(0.009)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,301	3,256	1,379	2,714	2,714	2,714	2,714
Number of MA	110	110	93	109	109	109	109
R^2	0.040	0.031	0.089	0.083	0.081	0.055	0.048

Notes: (i) In parentheses: standard errors clustered at the metropolitan area level; (ii) Treatment stands for "average treatment"; Wage is mean hourly wage and P90 is the ninetieth percentile of hourly wage. The estimates correspond to the coefficients β and λ in equation (4); (iii) In order to ease the interpretation of the coefficient associated with the interaction of treatment and geography, the global index of spatial isolation has been standardized according to its distribution among treated IRIS.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Sources: SIRENE, DADS, BD-TOPO and GIS SG-CIV

Table D3—Severance, Accessibility, Centrality and Unobserved Geographic Heterogeneity

	Inflows (1)	Creations (2)	Transfers (3)	Jobs (4)	Hours (5)	Wage (6)	P90 (7)
Treatment	0.044	−0.021	0.019	0.054*	0.029	0.016*	0.025*
	(0.032)	(0.037)	(0.068)	(0.031)	(0.036)	(0.0084)	(0.013)
Treat. × Sev.	−0.063**	−0.041	−0.108	−0.025	−0.0058	−0.012**	−0.024***
	(0.027)	(0.031)	(0.066)	(0.036)	(0.034)	(0.0057)	(0.007)
Treat. × Acc.	0.059***	0.065**	0.035	0.0047	0.023	−0.0015	−0.001
	(0.020)	(0.027)	(0.066)	(0.025)	(0.028)	(0.0050)	(0.008)
Treat. × Centr.	0.112***	0.093***	−0.096	0.054	0.073*	−0.010	−0.016
	(0.025)	(0.029)	(0.068)	(0.038)	(0.039)	(0.0067)	(0.012)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,301	3,256	1,379	2,714	2,714	2,714	2,714
Number of MA	110	110	93	109	109	109	109
R^2	0.040	0.031	0.090	0.084	0.081	0.056	0.049

Notes: (i) In parentheses: standard errors clustered at the metropolitan area level; (ii) Treatment stands for "average treatment"; Wage is mean hourly wage and P90 is the ninetieth percentile of hourly wage. The estimates correspond to the coefficients β and λ_k in equation (6).

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Sources: SIRENE, DADS, BD-TOPO and GIS SG-CIV

REFERENCES

Bartik, Timothy J. 2004. "Evaluating the Impacts of Local Economic Development Policies on Local Economic Outcomes: What has been done and what is doable?" In *Evaluating Local Economic and Employment Development: How to Access What Works Among Programmes and Policies*, 113–42. Paris: Organisation for Economic Co-Operation and Development (OECD).

Behaghel, Luc, Adrien Lorenceau, and Simon Quantin. 2013. "Replacing Churches and Mason Lodges? Tax Exemptions and Rural Development." Institute for the Study of Labor (IZA) Discussion Paper 7808.

Bondonio, Daniele, and John Engberg. 2000. "Enterprise Zones and Local Employment: Evidence from the States' Programs." *Regional Science and Urban Economics* 30 (5): 519–49.

Bondonio, Daniele, and Robert T. Greenbaum. 2007. "Do Local Tax Incentives Affect Economic Growth? What Mean Impacts Miss in the Analysis of Enterprise Zones Policies." *Regional Science and Urban Economics* 37 (1): 121–36.

Briant, Anthony, Pierre-Philippe Combes, and Miren Lafourcade. 2010. "Dots to Boxes: Do the Size and Shape of Spatial Units Jeopardize Economic Geography Estimations?" *Journal of Urban Economics* 67 (3): 287–302.

Briant, Anthony, Miren Lafourcade, and Benoît Schmutz. 2012. "Can Tax Breaks Beat Geography? Lessons from the French Enterprise Zone Program." Paris School of Economics CPSE/Working Paper 2012–22.

Briant, Anthony, Miren Lafourcade, and Benoît Schmutz. 2015. "Can Tax Breaks Beat Geography? Lessons from the French Enterprise Zone Experience: Dataset." *American Economic Journal: Economic Policy*. http://dx.doi.org/10.1257/pol.20120137.

Busso, Matias, Jesse Gregory, and Patrick Kline. 2013. "Assessing the Incidence and Efficiency of a Prominent Place Based Policy." *American Economic Review* 103 (2): 897–947.

Button, Kenneth. 2010. *Transport Economics.* Cheltenham, UK: Edward Elgar.

Charnoz, Pauline. 2014. "Do place-based policies help people? Evidence from French enterprise zones." Centre de Recherche en Économie et Statistique (CREST) Working Paper 2014–44.

Coulson, N. Edward, Derek Laing, and Ping Wang. 2001. "Spatial Mismatch in Search Equilibrium." *Journal of Labor Economics* 19 (4): 949–72.

Deaton, Angus. 2010. "Instruments, Randomization, and Learning about Development." *Journal of Economic Literature* 48 (2): 424–55.

Donzelot, Jacques, Catherine Mével, and Anne Wyvekens. 2003. "Faire société. La politique de la ville aux *États-Unis* et en France." *Revue française de pédagogie* 148 (148): 181–86.

Freedman, Matthew. 2012. "Teaching New Markets Old Tricks: The Effects of Subsidized Investment on Low-Income Neighborhoods." *Journal of Public Economics* 96 (11–12): 1000–1014.

Freedman, Matthew. 2013. "Targeted Business Incentives and Local Labor Markets." *Journal of Human Resources* 48 (2): 311–44.

Givord, Pauline, Simon Quantin, and Corentin Trevien. 2012. "A Long-Term Evaluation of the First Generation of the French Urban Enterprise Zones." Institut National de la Statistique et des Études Économiques (INSEE) Working Paper g2012-01.

Givord, Pauline, Roland Rathelot, and Patrick Sillard. 2013. "Place-Based Tax Exemptions and Displacement Effects: An Evaluation of the *Zones Franches Urbaines* Program." *Regional Science and Urban Economics* 43 (1): 151–63.

Glaeser, Edward, and Joshua Gottlieb. 2008. "The Economics of Place-Making Policies." *Brookings Papers on Economic Activity* 38 (1): 155–253.

Gobillon, Laurent, Thierry Magnac, and Harris Selod. 2012. "Do Unemployed Workers Benefit from Enterprise Zones? The French Experience." *Journal of Public Economics* 96 (9–10): 881–92.

Ham, John C., Charles Swenson, Ayse Imrohoroglu, and Heonjae Song. 2011. "Government programs can improve local labor markets: Evidence from State Enterprise Zones, Federal Empowerment Zones and Federal Enterprise Community." *Journal of Public Economics* 95 (7–8): 779–97.

Handy, Susan. 2003. "Amenity and Severance." In *Handbook of Transport and the Environment*, edited by David A. Hensher and Kenneth J. Button, 117–40. Oxford: Elsevier Science.

Harris, Chauncy D. 1954. "The Market As A Factor In The Localization Of Industry In The United States." *Annals of the Association of American Geographers* 44 (4): 315–48.

Jacobs, Jane. 1961. *The Death and Life of Great American Cities.* New York: Random House.

Kline, Patrick. 2010. "Place Based Policies, Heterogeneity, and Agglomeration." *American Economic Review* 100 (2): 383–87.

Kline, Patrick, and Enrico Moretti. 2013. "Place Based Policies with Unemployment." *American Economic Review* 103 (3): 238–43.

Kolko, Jed, and David Neumark. 2010. "Do some enterprise zones create jobs?" *Journal of Policy Analysis and Management* 29 (1): 5–38.

Landers, Jim. 2006. "Why Don't Enterprise Zones Work? Estimates of the Extent that EZ Benefits are Capitalized into Property Values." *Journal of Regional Analysis and Policy* 36 (1): 15–30.

Lynch, Devon, and Jeffrey S. Zax. 2011. "Incidence and substitution in enterprise zone programs: The case of Colorado." *Public Finance Review* 39 (2): 226–55.

Mayer, Thierry, Florian Mayneris, and Loriane Py. 2012. "The Impact of Urban Enterprise Zones on Establishment Location Decisions—Evidence from French ZFUs." Center for Economic and Policy Research (CEPR) Discussion Paper 9074.

Neumark, David, and Jed Kolko. 2010. "Do enterprise zones create jobs? Evidence from California's enterprise zone program." *Journal of Urban Economics* 68 (1): 1–19.

Neumark, David, and Helen Simpson. 2014. "Place-Based Policies." National Bureau of Economic Research (NBER) Discussion Paper 20049.

O'Keefe, Suzanne. 2004. "Job creation in California's enterprise zones: A comparison utilizing a propensity score matching model." *Journal of Urban Economics* 55 (1): 131–50.

Patacchini, Eleonora, and Yves Zenou. 2007. "Spatial Dependences in Local Unemployment Rates." *Journal of Economic Geography* 7 (2): 169–91.

Quantin, Simon. 2012. "Les embauches exonérées dans les territoires de la politique de la ville de 2007 à 2010: Forte baisse des embauches, en zone franche urbaine comme en zone de redynamisation urbaine." *Dares Analyses-Dares Indicateurs* 39: 1–9.

Rathelot, Roland, and Patrick Sillard. 2007. "Zones Franches Urbaines: quels effets sur l'emploi salarié et les créations d'établissements?" Institut National de la Statistique et des Etudes Economiques (DESE) Working Paper g2007-11.

Rathelot, Roland, and Patrick Sillard. 2008. "Zones Franches Urbaines: quels effets sur l'emploi salarié et les créations d'établissements?" *Economie et Statistique* 415–16: 81–96.

Rogers, Cynthia L. 1997. "Job Search and Unemployment Duration: Implications for the Spatial Mismatch Hypothesis." *Journal of Urban Economics* 42 (1): 109–32.

Wang, Jin. 2013. "The Economic Impact of Special Economic Zones: Evidence from Chinese Municipalities." *Journal of Development Economics* 101: 133–47.

Zenou, Yves. 2002. "How Do Firms Redline Workers?" *Journal of Urban Economics* 52 (3): 391–408.